Chapter 6
Aircraft Weight & Balance

Introduction

The weight of an aircraft and its balance are extremely important for operating in a safe and efficient manner. When a manufacturer designs an aircraft and the Federal Aviation Administration (FAA) certifies it, the specifications identify the aircraft's maximum weight and the limits within which it must balance. The weight and balance system commonly employed among aircraft consists of three equally important elements: the weighing of the aircraft, the maintaining of the weight and balance records, and the proper loading of the aircraft.

The maximum weight of an aircraft is based on the amount of lift the wings or rotors can provide under the operating conditions for which the aircraft is designed. For example, if a small general aviation (GA) airplane required a takeoff speed of 200 miles per hour (mph) to generate enough lift to support its weight, that would not be safe. Taking off and landing at lower airspeeds is certainly safer than doing so at higher speeds.

Aircraft balance is also a significant factor in determining if the aircraft is safe to operate. An aircraft that does not have good balance can exhibit poor maneuverability and controllability, making it difficult or impossible to fly. This could result in an accident, causing damage to the aircraft and injury to the people on board. Safety is the primary reason for concern about an aircraft's weight and balance.

Another important reason for concern about weight and balance is the efficiency of the aircraft. Improper loading reduces the efficiency of an aircraft from the standpoint of ceiling, maneuverability, rate of climb, speed, and fuel consumption. If an airplane is loaded in such a way that it is extremely nose heavy, higher than normal forces are exerted at the tail to keep the airplane in level flight. The higher than normal forces at the tail create additional drag, which requires additional engine power and therefore additional fuel flow to maintain airspeed.

The most efficient condition for an aircraft is to have the point where it balances fall close to, or exactly at, the aircraft's center of lift. If this were the case, little or no flight control force would be needed to keep the aircraft flying straight and level. In terms of stability and safety, however, this perfectly balanced condition might not be desirable. All factors that affect aircraft safety and efficiency, in terms of its weight and balance, are discussed in detail in this chapter.

Requirements for Aircraft Weighing

Every aircraft type certificated by the FAA receives a weight and balance report as part of its required aircraft records before leaving the factory for delivery to its new owner. The weight and balance report identifies the empty weight of the aircraft and the location at which the aircraft balances, known as the center of gravity (CG). The weight and balance report must include an equipment list showing weights and moment arms of all required and optional items of equipment included in the certificated empty weight. If the manufacturer chooses to do so, it can weigh every aircraft it produces and issue the weight and balance report based on that weighing. As an alternative, the manufacturer is permitted to weigh an agreed upon percentage of a particular model of aircraft produced, perhaps 10 to 20 percent, and apply the average to all the aircraft.

After the aircraft leaves the factory and is delivered to its owner, the requirement for placing the aircraft on scales and reweighing it varies depending on the type of aircraft and how it is used. For a small, GA airplane being used privately, such as a Cessna 172, there is no FAA requirement that it be periodically reweighed; but after each annual, the mechanic must ensure that the weight and balance data in the aircraft records is correct. Additionally, there is an FAA requirement that the airplane always have a current and accurate weight and balance report. If the weight and balance report for an aircraft is lost, the aircraft must be weighed and a new report must be created. When an aircraft has undergone extensive repair, major alteration, or has new equipment installed, such as a radio or a global positioning system, a new weight and balance report must be created. The equipment installer may place the airplane on scales and weigh it after the installation, which is an acceptable way of creating the new report. If the installer knows the exact weight and location of the new equipment, it is also possible to create a new report by doing a series of mathematical calculations.

Over time, almost all aircraft tend to gain weight. Examples of how this can happen include an airplane being repainted without the old paint being removed and the accumulation of

dirt, grease, and oil in parts of the aircraft that are not easily accessible for cleaning. When new equipment is installed, and its weight and location are mathematically accounted for, some miscellaneous weight might be overlooked, such as wire and hardware. For this reason, even if the FAA does not require it, it is a good practice to periodically place an aircraft on scales and confirm its actual empty weight and empty weight center of gravity (EWCG).

Some aircraft are required to be weighed and have their CG calculated on a periodic basis, typically every 3 years. Examples of aircraft that fall under this requirement are:

1. Air taxi and charter twin-engine airplanes operating under Title 14 of the Code of Federal Regulations (14 CFR) part 135, section 135.185(a).
2. Airplanes with a seating capacity of 20 or more passengers or a maximum payload of 6,000 pounds or more, as identified in 14 CFR part 125, section 125.91(b).

Weight & Balance Terminology

Datum

The datum is an imaginary vertical plane from which all horizontal measurements are taken for balance purposes, with the aircraft in level flight attitude. If the datum is viewed on a drawing of an aircraft, it would appear as a vertical line that is perpendicular (90 degrees) to the aircraft's longitudinal axis. For each aircraft make and model, the location of all items is identified in reference to the datum. For example, the fuel in a tank might be 60 inches (60") behind the datum, and a radio on the flight deck might be 90" forward of the datum.

The datum is determined by the manufacturer; it is often the leading edge of the wing or some specific distance from an easily identified location. Typical locations for the datum are the aircraft nose, the leading edge of the wing, the helicopter's mast, or a specified distance from a known point. However, most modern helicopters, like airplanes, have the datum located at the nose of the aircraft or a specified distance ahead of it. *Figure 6-1* shows an aircraft with the leading edge of the wing being the datum. The distance from this datum is measured in inches and can be either positive or negative depending upon where the equipment is located in relation to the datum.

The location of the datum is identified in the Aircraft Specifications or Type Certificate Data Sheet (TCDS). Aircraft certified prior to 1958 fell under the Civil Aeronautics Administration and had their weight and balance information contained in a document known as Aircraft Specifications. Aircraft certified since 1958 fall under the FAA and have their weight and balance information contained in a document known as a Type Certificate Data Sheet (TCDS). The Aircraft Specifications typically included the aircraft equipment list. For aircraft with a TCDS, the equipment list is a separate document.

Arm

The arm is the horizontal distance from the datum to any point within the aircraft. The arm's distance is always measured in inches, and it is preceded by the algebraic sign for positive (+) or negative (−), except for a location which might be exactly on the datum. The positive sign indicates an item is located aft of the datum, and the negative sign indicates an item is located forward of the datum. If the manufacturer chooses a datum that is at the most forward location on an aircraft, all the arms will be positive numbers. Location of the datum at any other point on the aircraft results in some arms being positive numbers, or aft of the datum, and some arms being negative numbers, or forward of the datum. *Figure 6-1* shows an aircraft where the datum is the leading edge of the wing. For this aircraft, any item (fuel, seat, radio, etc.) located forward of the wing leading edge has a negative arm, and any item located aft of the wing leading edge has a positive arm. If an item is located exactly at the wing leading edge, its arm would be zero, and mathematically it would not matter whether its arm was positive or negative.

The arm of each item is usually included in parentheses immediately after the item's name or weight in the Aircraft Specifications, TCDS, or equipment list for the aircraft. For example, in a TCDS, the fuel quantity might be identified as 98 gallons (gal) (+93.6) and the forward baggage limit as 100 pounds (lb) (–22.5). These numbers indicate that the fuel is located 93.6" aft of the datum and the nose baggage is located 22.6" forward of the datum. If the arm for a piece of equipment is not known, its exact location must be accurately measured. When the arm for a piece of equipment is being determined, the measurement is taken from the datum to the piece of equipment's own CG.

Moment

To understand balance, it is necessary to have a working knowledge of the principle of moments. For those unfamiliar with weight and balance terms, the word moment is the product of a force or weight times a distance. The distance used in calculating a moment is referred to as the arm or moment arm and is usually expressed in inches. To calculate a moment, a force (or weight) and a distance must be known. The weight is multiplied by the distance from the datum and the result is the moment, which is expressed in inch-pounds (in-lb), a point through which the force acts. For the purpose of illustration, compare an aircraft to a seesaw. Like the seesaw, for an aircraft to be in balance, or equilibrium, the

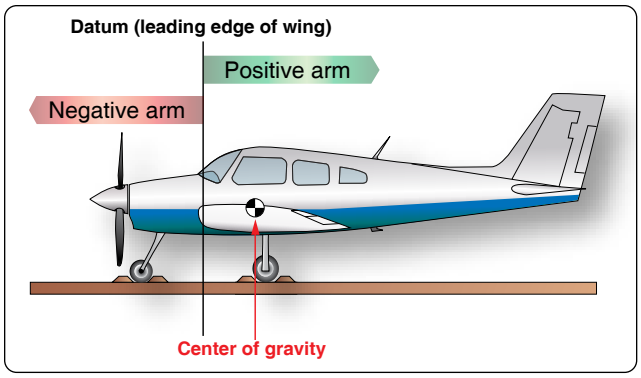


Figure 6-1. *Datum location and its effect on positive and negative arms.*

sum of the moments on each side of the balance point must be equal. Therefore, the same weight that is different distances (in inches) from the datum have greater moments.

A 5 lb radio located 80" from the datum would have a moment of 400 in-lb (5 lb × 80"). A 10-pound radio located 12" from the datum would have a moment of 120 in-lb. Whether the moment is preceded by a positive (+) or negative (−) sign depends on its location in relation to the datum. *Figure 6-2* shows where the moment ends up being a positive number because the weight and arm are both positive.
The algebraic sign of the moment, based on the datum location and whether weight is being installed or removed *[Figure 6-3]*, would be as follows:

- Weight being added aft of the datum produces a positive moment (+weight, +arm).
- Weight being added forward of the datum produces a negative moment (+weight, −arm).
- Weight being removed aft of the datum produces a negative moment (−weight, +arm).
- Weight being removed forward of the datum produces a positive moment (−weight, −arm).

When dealing with positive and negative numbers, remember that the product of like signs produces a positive answer, and the product of unlike signs produces a negative answer.

Center of Gravity (CG)

The CG is the point at which all the weight of the aircraft is concentrated and balanced; therefore, the aircraft can be supported at that point (the CG). The magnitude of the nose-heavy and tail-heavy moments are exactly equal. It is the balance point for the aircraft and, if suspended from this point, there would be no tendency to rotate in a noseup or nosedown attitude.

Figure 6-4 shows a lever with the pivot point (called a fulcrum) located at the CG for the lever. Even though the weights on either side of the fulcrum are not equal, and the distances from each weight to the fulcrum are not equal, the product of the weights and arms (moments) are equal, and that is what produces a balanced condition. Therefore, the lever would be balanced much like two persons sitting on a seesaw who are differing weights and located at different distances from the fulcrum.

Maximum Weight

The maximum weight is the maximum authorized weight of the aircraft and its contents, and is indicated in the Aircraft Specifications or TCDS. For many aircraft, there are variations to the maximum allowable weight depending on the purpose and conditions under which the aircraft is to be flown. For example, a certain aircraft may be allowed a maximum gross weight of 2,750 lb when flown in the normal category, but when flown in the utility category, which allows for limited aerobatics, the same aircraft's maximum allowable gross weight might only be 2,175 lb. There are other variations when dealing with the concept of maximum weight, as follows:

- Maximum Ramp Weight—the heaviest weight to which an aircraft can be loaded while it is sitting on the ground. This is sometimes referred to as the maximum taxi weight.
- Maximum Takeoff Weight—the heaviest weight an aircraft can be when it starts the takeoff roll. The difference between this weight and the maximum ramp weight would equal the weight of the fuel that would be consumed prior to takeoff.
- Maximum Landing Weight—the heaviest weight an aircraft can be when it lands. For large, wide body commercial airplanes, it can be 100,000 lb less than maximum takeoff weight, or even more.
- Maximum Zero Fuel Weight—the heaviest weight an aircraft can be loaded to without having any usable fuel in the fuel tanks. Any weight loaded above this value must be in the form of fuel.

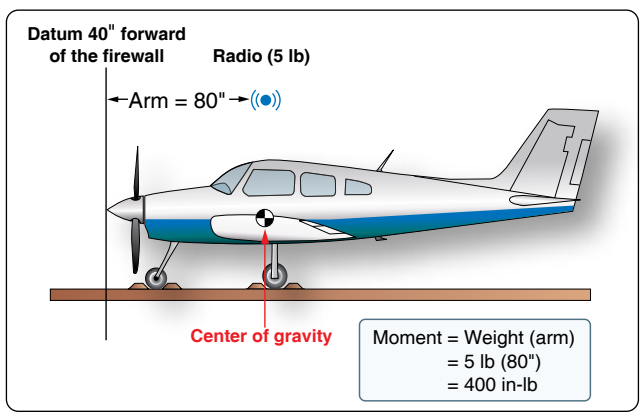


Figure 6-2. *Moment of a radio located aft of the datum.*

Empty Weight

The empty weight of an aircraft includes all operating equipment that has a fixed location and is actually installed in the aircraft. It includes the weight of the airframe, powerplant, required equipment, optional or special equipment, fixed ballast, hydraulic fluid, and residual fuel and oil. Residual fuel and oil are the fluids that do not normally drain out because they are trapped in the fuel lines, oil lines, and tanks. They must be included in the aircraft's empty weight. For most aircraft certified after 1978, the full capacity of the engine oil system is also included in the empty weight. Information regarding residual fluids in aircraft systems that must be included in the empty weight, and whether or not full oil is included, will be indicated in the Aircraft Specifications or TCDS.

Other terms that are used when describing empty weight include basic empty weight, licensed empty weight, and standard empty weight. The term "basic empty weight" applies when the full capacity of the engine oil system is included in the value. The term "licensed empty weight" applies when only the weight of residual oil is included in the value, so it generally involves only aircraft certified prior to 1978. Standard empty weight would be a value supplied by the aircraft manufacturer, and it would not include any optional equipment that might be installed in an aircraft. For most people working in the aviation maintenance field, the basic empty weight of the aircraft is the most important one.

Empty Weight Center of Gravity (EWCG)

The EWCG for an aircraft is the point at which it balances when it is in an empty weight condition. The concepts of empty weight and CG were discussed earlier in this chapter, and now they are being combined into a single concept.

One of the most important reasons for weighing an aircraft is to determine its EWCG. All other weight and balance calculations, including loading the aircraft for flight, performing an equipment change calculation, and performing an adverse condition check, begin with knowing the empty weight and EWCG. This crucial information is part of what is contained in the aircraft weight and balance report.

Weight	Arm	Moment	Rotation
+	+	+	Noseup
+	–	–	Nosedown
–	+	–	Nosedown
–	–	+	Noseup

Figure 6-3. *Relationship between the algebraic signs of weight, arms, and moments.*

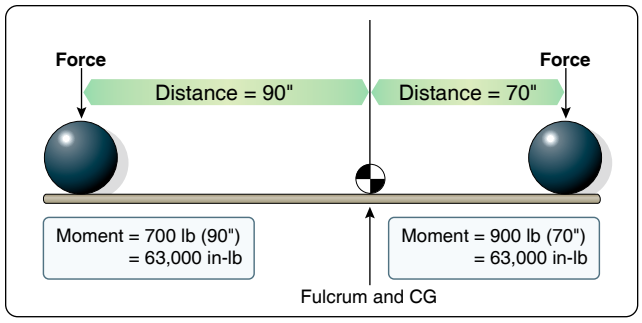


Figure 6-4. *Center of gravity and a first class lever.*

Useful Load

To determine the useful load of an aircraft, subtract the empty weight from the maximum allowable gross weight. For aircraft certificated in both normal and utility categories, there may be two useful loads listed in the aircraft weight and balance records. An aircraft with an empty weight of 3,100 lb may have a useful load of 850 lb, if the normal category maximum weight is listed as 3,950 lb. When the aircraft is operated in the utility category, the maximum gross weight may be reduced to 3,700 lb, with a corresponding decrease in the useful load to 600 lb. Some aircraft have the same useful load regardless of the category in which they are certificated.

The useful load consists of fuel, any other fluids that are not part of empty weight, passengers, baggage, pilot, copilot, and crewmembers. Whether the weight of engine oil is considered part of the useful load depends on when the aircraft was certificated and can be determined by looking at the Aircraft Specifications or TCDS. The payload of an aircraft is like the useful load, except it does not include fuel.

A reduction in the weight of an item, where possible, may be necessary to remain within the maximum weight allowed for the category in which an aircraft is operating. Determining the distribution of these weights is called a weight check.

Minimum Fuel

Many modern aircraft have multiple rows of seats and often more than one baggage compartment. The weight and balance extreme conditions represent the maximum forward and rearward CG position for the aircraft. An aircraft has certain fixed points, fore and aft, beyond which the CG should not be permitted at any time during flight. A check should be made to ensure that the CG will not shift out of limits when crew, passengers, cargo, and expendable weights are added or removed. If the limits are exceeded and the aircraft is flown in this condition, it may lead to insufficient stability, with resulting difficulty in controlling the aircraft. After any repair or alteration that changes the weight and balance, the Airframe and Powerplant (A&P) mechanic or repairman

must ensure that no legal condition of loading can move the CG outside of its allowable limits. To determine this, the mechanic will deliberately attempt to calculate the aircraft loading in such a manner as to place the CG outside the limits of the aircraft. This is called an adverse-loading check.

For example, in a forward adverse-loaded CG check, all useful load in front of the forward CG limit is loaded, and all useful load behind this limit is left empty. An exception to leaving it empty is the fuel tank. If the fuel tank is located behind the forward CG limit, it cannot be left empty because the aircraft cannot fly without fuel. In this case, an amount of fuel is accounted for, which is known as minimum fuel. Minimum fuel is the amount needed for 30 minutes of flight at cruise power.

For weight and balance purposes, the minimum fuel is no more than the quantity needed for one half hour of operation at rated maximum continuous power. This is 1/12 gallon for each maximum except takeoff (METO) horsepower (hp). Because aviation gasoline (Avgas) weighs 6 pounds per gallon (lb/gal), determine the number of pounds of the minimum fuel by dividing the METO hp by 2. For instance, an aircraft having a METO hp of 200 hp will have a minimum fuel of 16.65 gallons or 99.99 pounds. An even simpler way is to take the METO hp divided by 2, which is 100 pounds. Both methods in determining minimum fuel are valued and result in essentially the same answer. In the latter computation, a piston engine in cruise flight burns 1 lb of fuel per hour for each hp, or ½ lb for 30 minutes, hence dividing the METO by 2.

For example, if a forward adverse-loaded CG check was performed on a piston engine aircraft, with the engine having a METO hp of 200, the minimum fuel would be 100 lb (200 METO hp ÷ 2).

For turbine engine-powered aircraft, minimum fuel is not based on engine hp. If an adverse-loaded CG check is being performed on a turbine engine-powered aircraft, the aircraft manufacturer would need to supply information on minimum fuel.

Tare Weight

When aircraft are placed on scales and weighed, it is sometimes necessary to use support equipment to aid in the weighing process. For example, to weigh a tail dragger airplane, it is necessary to raise the tail to get the airplane level. To level the airplane, a jack might be placed on the scale and used to raise the tail. Unfortunately, the scale is now absorbing the weight of the jack in addition to the weight of the airplane. This extra weight is known as tare weight and must be subtracted from the scale reading. Other examples of tare weight are wheel chocks placed on the scales and ground locks left in place on retractable landing gear.

Procedures for Weighing an Aircraft

General Concepts

The most important reason for weighing an aircraft is to find out its empty weight (basic empty weight) and to find out where it balances in the empty weight condition. When an aircraft is to be flown, the pilot-in-command must know what the loaded weight of the aircraft is and where its loaded CG is. For the loaded weight and CG to be calculated, the pilot or dispatcher handling the flight must first know the empty weight and EWCG.

Earlier in this chapter it was identified that the CG for an object is the point about which the nose heavy and tail heavy moments are equal. One method that could be used to find this point would involve lifting an object off the ground twice, first suspending it from a point near the front, and on the second lift suspending it from a point near the back. With each lift, a perpendicular line (90 degrees) would be drawn from the suspension point to the ground. The two perpendicular lines would intersect somewhere in the object, and the point of intersection would be the CG. This concept is shown in *Figure 6-5*, where an airplane is suspended from two different points. The perpendicular line from the first suspension point is shown in red, and the new suspension point line is shown as a blue plumb bob. Where the red and blue lines intersect is the CG. If an airplane were suspended from two points, one at the nose and one at the tail, the perpendicular drop lines would intersect at the CG. Suspending an airplane from the ceiling by two hooks, however, is clearly not realistic. Even if it could be done, determining where in the airplane the lines intersect would be difficult.

A more realistic way to find the CG for an object, especially an airplane, is to place it on a minimum of two scales and calculate the moment value for each scale reading. In *Figure 6-6,* there is a plank that is 200" long, with the left end being the datum (zero arm), and 6 weights placed at various locations along the length of the plank. The purpose of *Figure 6-6* is to show how the CG can be calculated when the arms and weights for an object are known.

To calculate the CG for the object in *Figure 6-6,* the moments for all the weights need to be calculated and then summed, and the weights need to be summed. In the four-column table in *Figure 6-7,* the item, weight, and arm are listed in the first three columns, with the information coming from *Figure 6-6.* The moment value in the fourth column is the product of the weight and arm. The weight and moment

columns are summed, with the CG being equal to the total moment divided by the total weight. The arm column is not summed. The number appearing at the bottom of that column is the CG. The calculation is shown in *Figure 6-7.*

For the calculation in *Figure 6-7,* the total moment is 52,900 in-lb, and the total weight is 495 lb. The CG is calculated as follows:

CG = Total Moment ÷ Total Weight
= 52,900 in-lb ÷ 495 lb
= 106.9" (106.87 rounded to tenths)

An interesting characteristic exists for the problem in *Figure 6-6* and the table showing the CG calculation. If the datum (zero arm) for the object was in the middle of the 200" long plank, with 100" of negative arm to the left and 100" of positive arm to the right, the solution would show the CG to be in the same location. The arm for the CG would not be the same number, but its physical location would be the same. *Figures 6-8* and *6-9* show the new calculation.

CG = Total Moment ÷ Total Weight
= 3,400 in-lb ÷ 495 lb
= 6.9" (6.87 rounded to tenths)

In *Figure 6-8,* the CG is 6.9" to the right of the plank's center. Even though the arm is not the same number, in *Figure 6-6* the CG is also 6.9" to the right of center (CG location of 106.9 with the center being 100). Because both problems are the same in these two figures, except for the datum location, the CG must be the same.

The definition for CG states that it is the point about which all the moments are equal. We can prove that the CG for the object in *Figure 6-8* is correct by showing that the total moments on either side of this point are equal. Using 6.87 as the CG location for slightly greater accuracy, instead of the rounded off 6.9 number, the moments to the left of the CG are shown in *Figure 6-10.* The moments to the right of the CG, shown in *Figure 6-8,* would be as indicated in *Figure 6-11.* Disregarding the slightly different decimal value, the moment in both previous calculations is 10,651 in-lb. Showing that the moments are equal is a good way of proving that the CG has been properly calculated.

Weight and Balance Data

Before an aircraft can be properly weighed and its EWCG computed, certain information must be known. This information is furnished by the FAA to anyone for every certificated aircraft in the TCDS or Aircraft Specifications. When the design of an aircraft is approved by the FAA, an Approved Type Certificate and TCDS are issued. The TCDS

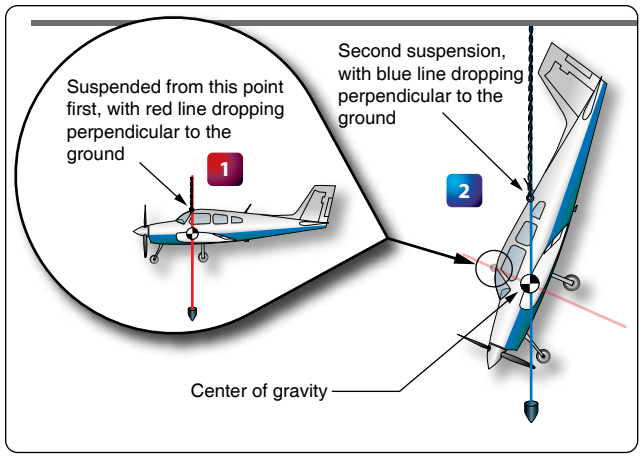


Figure 6-5. *Center of gravity determined by two suspension points.*

includes all the pertinent specifications for the aircraft, and at each annual or 100-hour inspection, it is the responsibility of the inspecting mechanic or repairman to ensure that the aircraft adheres to them.

Manufacturer-Furnished Information

When an aircraft is initially certificated, its empty weight and EWCG are determined and recorded in the weight and balance record, such as the one in *Figure 6-12.* An equipment list is furnished with the aircraft that specifies all the required equipment and all equipment approved for installation in the aircraft. The weight and arm of each item is included on the

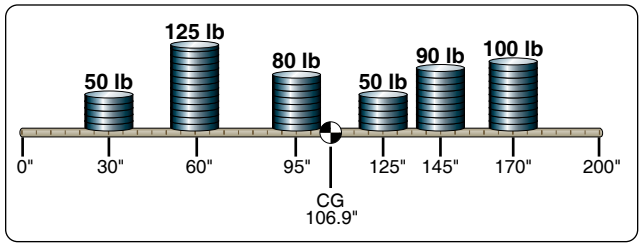


Figure 6-6. *Center of gravity for weights on a plank with datum at one end.*

Item	Weight (lb)	×	Arm (inches)	=	Moment (in-lb)
50 pound weight	50		+30		1,500
125 pound weight	125		+60		7,500
80 pound weight	80		+95		7,600
50 pound weight	50		+125		6,250
90 pound weight	90		+145		13,050
100 pound weight	100		+170		17,000
Total	**495**		**+106.9**		**52,900**

Figure 6-7. *Center of gravity calculation for weights on a plank with datum at one end.*

list, and all equipment installed when the aircraft left the factory is checked. When an aircraft mechanic or repairman adds or removes any item on the equipment list, they must change the weight and balance record to indicate the new empty weight and EWCG, and the equipment list is revised to show what is installed.

Figure 6-13 is an excerpt from a comprehensive equipment list that includes all the items of equipment approved for this model of aircraft. The Pilot's Operating Handbook (POH) or Airplane Flight Manual (AFM) for each individual aircraft includes an aircraft specific equipment list of the items from this master list. When any item is added to or removed from the aircraft, its weight and arm are determined in the equipment list and used to update the weight and balance record. The POH/AFM also contains CG moment envelopes and loading graphs.

Figures 6-14 through *6-16* shows a TCDS for a Piper twin-engine airplane known as the Seneca (PA-34-200). The main headings for the information contained in a TCDS are included, but much of the information contained under these headings has been removed if it did not directly pertain to weight and balance. Information on only one model of Seneca is shown, because to show all the different models would make the document excessively long. The portion of the TCDS that has the most direct application to weight and balance is highlighted in yellow.

Some of the important weight and balance information found in a TCDS is as follows:

1. Engine
2. CG range
3. Maximum weight
4. Number of seats
5. Baggage capacity
6. Fuel capacity
7. Oil capacity
8. Datum information
9. Leveling means

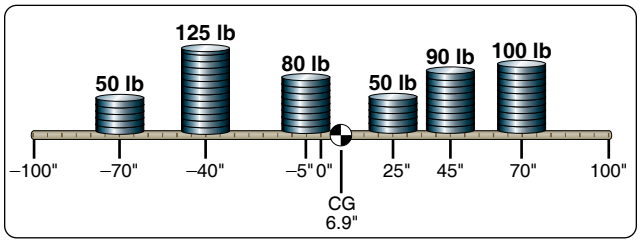


Figure 6-8. *Center of gravity for weights on a plank with datum in the middle.*

Item	Weight (lb)	×	Arm (inches)	=	Moment (in-lb)
50 pound weight	50		–70		–3,500
125 pound weight	125		–40		–5,000
80 pound weight	80		–5		–400
50 pound weight	50		+25		1,250
90 pound weight	90		+45		4,050
100 pound weight	100		+70		7,000
Total	**495**		**+6.9**		**3,400**

Figure 6-9. *Center of gravity calculation for weights on a plank with datum in the middle.*

Item	Weight (lb)	×	Arm (inches)	=	Moment (in-lb)
50 pound weight	50		76.87		3,843.50
125 pound weight	125		46.87		5,858.75
80 pound weight	80		11.87		949.60
Total	**255**		**135.61**		**10,651.85**

Figure 6-10. *Moments to the left of the center of gravity.*

10. Amount of oil in empty weight
11. Amount of fuel in empty weight

Weight and Balance Equipment

Scales

Weighing GA aircraft, helicopters, turboprops, corporate jets, UAV/UAS, or transport category airliners can be accomplished in two ways: top of jack load cells and platform scales. Equipment selection is dependent on the operator's needs and or equipment currently on hand, as well as the airframe manufacturer's recommendations. Top of jack load cells, as the name implies, can be used on top of the current wing jacks or can be used under axle for larger jets. Platforms are very useful for small shops that do not have jacks for every type of aircraft.

Both types of scales feature new technologies using wireless

Item	Weight (lb)	×	Arm (inches)	=	Moment (in-lb)
50 pound weight	50		18.13		906.50
90 pound weight	90		38.13		3,431.70
100 pound weight	100		63.13		6,313.00
Total	**240**		**119.39**		**10,651.25**

Figure 6-11. *Moments to the right of the center of gravity.*

Weight and Balance Data					
Aircraft Serial #: 18259080 FAA Registration #: N42565 Date: 04-22-2005					
Item	**Weight (lb)**	**×**	**CG Arm (in)**	**=**	**Moment (in-lb)**
Standard empty weight	1,876		36.1		67,798.6
Optional equipment	1.2		13.9		16.7
Special installation	6.2		41.5		257.3
Paint	–		–		–
Unusable fuel	30.0		46.0		1,380
Basic empty weight	1,913.4				69,452.6

Figure 6-12. *Typical weight and balance data for 14 CFR part 23 airplane.*

operations with computer-based indication and cable-based wired digital indication. Mechanical or analog meter scales have mostly been replaced with the new wireless systems and or digital indicators. These systems and indicators are very accurate and easy to use, making the weighing job faster to accomplish and providing higher quality in readings.

Platforms are available in many weight ranges and sizes. These systems either use ramps or the aircraft can be jacked and lowered onto the platforms during regular maintenance. Platforms are easy to use and are a choice for many shops that do not have jacks for the many types of aircraft to be serviced. The limiting factors for platforms are the weight range and the tire size, some aircraft have large tires and the platform may be too small for the specific aircraft tire. It is important to always use the right size scale and platform for the aircraft type and weighing job required.

The platform scale sits on the hangar floor in a level condition. Ramps and a tug are used to position the airplane tire on top of the platform and centered. Built into the platform is an electronic load cell(s) that sense the weight being applied to it, which generates a corresponding electrical signal. Inside the load cell is an electronic strain gauge that measures a proportional change in electrical resistance as the weight being applied to it increases. An electrical cable or wireless signal runs from the platform scale to a display unit, computer, or tablet, which interprets the resistance change of the load cell and equates it to a specific number of pounds. A digital readout on the display shows the weight. In *Figure 6-17,* a small Piper is being weighed using wireless platform scales that incorporate electronic load cells.

In *Figure 6-18,* a Cessna 182 airplane is being weighed with portable electronic platform scales. If an aircraft is weighed on platform scales, the only way to level the aircraft is to deflate tires and landing gear struts accordingly. This type of scale is easy to transport and can be powered by household current or by a battery contained in the display unit.

The display unit for the standard wired platform scales is very easy to use. *[Figure 6-19]* Turn on the power and the unit runs through the software and displays the scales in a total mode. Pressing on the ZERO KEY (blue key not the number key) will ZERO the channels. Once completed, the unit will read -0- and the scale is ready to use. Select the channels by number and pressing the PRINT/SELECT KEY. All channels can be returned to TOTAL MODE by entering the number 4 TOTAL followed by the PRINT/SELECT KEY. If all three scale switches are turned on at the same time, the total weight of the airplane is displayed.

The second type of aircraft scale is a top of jack, cell-based scale, where each jack point receives a cell-based transducer on the top of the jack. It is very easy to use and level the aircraft during the weighing operation. The system is easy to transport, light weight, and simple to set up. The operator must have a jack capable of receiving and mounting the cell. Cells come in many weight ranges and are dependent on the weight required per point to accomplish the weighing and receiving the actual jack point type.

The top of the load cell has a concave shape that matches up with the jack pad on the aircraft, with the load cell absorbing all the weight of the aircraft at each jacking point. Each load cell either has an electrical cable attached to it or is wireless, which connects to the display unit or computer read out that shows the weight transmitted to each load cell. An important advantage of weighing an aircraft this way is that it allows the technician to level the aircraft easily. When an aircraft

Comprehensive Equipment List

This is a comprehensive list of all Cessna equipment that is available for the Model 182S airplane. It should not be confused with the airplane-specific equipment list. An airplane-specific list is provided with each individual airplane at delivery and is typically inserted at the rear of this Pilot's Operating Handbook. The following comprehensive equipment list and the airplane-specific list have a similar order of listing.

The comprehensive equipment list provides the following information in column form:

In the **Item No** column, each item is assigned a coded number. The first two digits of the code represent the assignment of item within the ATA iSpec 2200 breakdown (Chapter 11 for Placards, Chapter 21 for Air Conditioning, Chapter 77 for Engine Indicating, etc.). These assignments also correspond to the Maintenance Manual chapter breakdown for the airplane. After the first two digits (and hyphen), items receive a unique sequence number (01, 02, 03, etc...). After the sequence number (and hyphen), a suffix letter is assigned to identify equipment as a required item, a standard item or an optional item. Suffix letters are as follows:

–R = required items or equipment for FAA certification
–S = standard equipment items
–O = optional equipment items replacing required or standard items
–A = optional equipment items which are in addition to required or standard items

In the **Equipment List Description** column, each item is assigned a descriptive name to help identify its function.

In the **Ref Drawing** column, a drawing number is provided which corresponds to the item.

Note

If additional equipment is to be installed, it must be done in accordance with the reference drawing, service bulletin or a separate FAA approval.

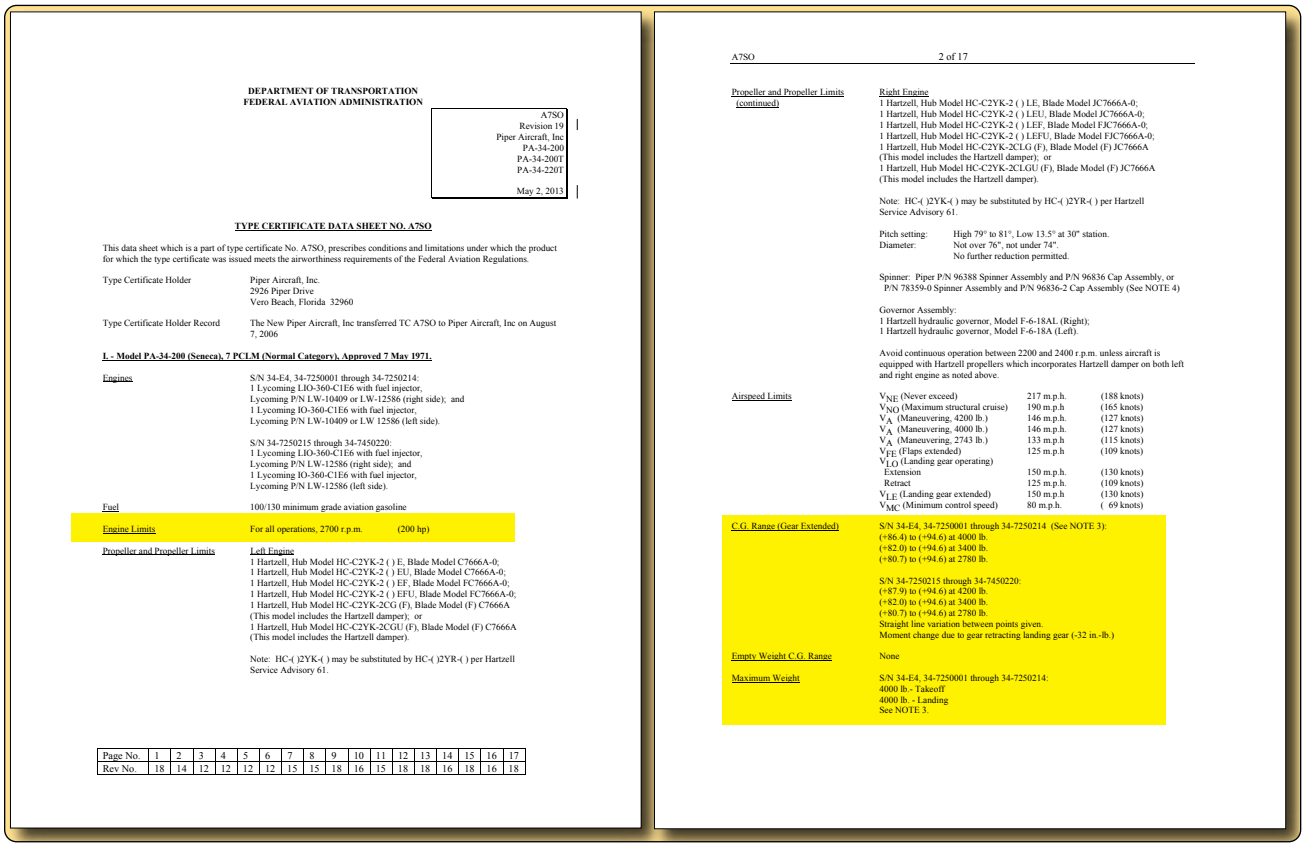
In the **Wt Lbs** and **Arm Ins** columns, information is provided on the weight (in pounds) and arm (in inches) of the equipment item.

Notes

Unless otherwise indicated, true values (not net change values) for the weight and arm are shown. Positive arms are distances aft of the airplane datum; negative arms are distances forward of the datum.

Asterisks (*) in the weight and arm column indicate complete assembly installations. Some major components of the assembly are listed on the lines immediately following. The sum of these major components does not necessarily equal the complete assembly installation.

Figure 6-13. *Excerpt from a typical comprehensive equipment list.*



DEPARTMENT OF TRANSPORTATION
FEDERAL AVIATION ADMINISTRATION

A7SO
Revision 19
Piper Aircraft, Inc
PA-34-200
PA-34-200T
PA-34-220T

May 2, 2013

TYPE CERTIFICATE DATA SHEET NO. A7SO

This data sheet which is a part of type certificate No. A7SO, prescribes conditions and limitations under which the product for which the type certificate was issued meets the airworthiness requirements of the Federal Aviation Regulations.

Type Certificate Holder — Piper Aircraft, Inc.
2926 Piper Drive
Vero Beach, Florida 32960

Type Certificate Holder Record — The New Piper Aircraft, Inc transferred TC A7SO to Piper Aircraft, Inc on August 7, 2006

I. - Model PA-34-200 (Seneca), 7 PCLM (Normal Category), Approved 7 May 1971.

Engines — S/N 34-E4, 34-7250001 through 34-7250214:
1 Lycoming LIO-360-C1E6 with fuel injector,
Lycoming P/N LW-10409 or LW-12586 (right side); and
1 Lycoming IO-360-C1E6 with fuel injector,
Lycoming P/N LW-10409 or LW 12586 (left side).

S/N 34-7250215 through 34-7450220:
1 Lycoming LIO-360-C1E6 with fuel injector,
Lycoming P/N LW-12586 (right side); and
1 Lycoming IO-360-C1E6 with fuel injector,
Lycoming P/N LW-12586 (left side).

Fuel — 100/130 minimum grade aviation gasoline

Engine Limits — For all operations, 2700 r.p.m. (200 hp)

Propeller and Propeller Limits — Left Engine
1 Hartzell, Hub Model HC-C2YK-2 ( ) E, Blade Model C7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) EU, Blade Model C7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) EF, Blade Model FC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) EFU, Blade Model FC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2CG (F), Blade Model (F) C7666A
(This model includes the Hartzell damper); or
1 Hartzell, Hub Model HC-C2YK-2CGU (F), Blade Model (F) C7666A
(This model includes the Hartzell damper).

Note: HC-( )2YK-( ) may be substituted by HC-( )2YR-( ) per Hartzell Service Advisory 61.

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Propeller and Propeller Limits (continued) — Right Engine
1 Hartzell, Hub Model HC-C2YK-2 ( ) LE, Blade Model JC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) LEU, Blade Model JC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) LEF, Blade Model FJC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) LEFU, Blade Model FJC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2CLG (F), Blade Model (F) JC7666A
(This model includes the Hartzell damper); or
1 Hartzell, Hub Model HC-C2YK-2CLGU (F), Blade Model (F) JC7666A
(This model includes the Hartzell damper).

Note: HC-( )2YK-( ) may be substituted by HC-( )2YR-( ) per Hartzell Service Advisory 61.

Pitch setting: High 79° to 81°, Low 13.5° at 30" station.
Diameter: Not over 76", not under 74".
No further reduction permitted.

Spinner: Piper P/N 96388 Spinner Assembly and P/N 96836 Cap Assembly, or P/N 78359-0 Spinner Assembly and P/N 96836-2 Cap Assembly (See NOTE 4)

Governor Assembly:
1 Hartzell hydraulic governor, Model F-6-18AL (Right);
1 Hartzell hydraulic governor, Model F-6-18A (Left).

Avoid continuous operation between 2200 and 2400 r.p.m. unless aircraft is equipped with Hartzell propellers which incorporates Hartzell damper on both left and right engine as noted above.

Airspeed Limits

V_{NE} (Never exceed)	217 m.p.h.	(188 knots)
V_{NO} (Maximum structural cruise)	190 m.p.h	(165 knots)
V_A (Maneuvering, 4200 lb.)	146 m.p.h.	(127 knots)
V_A (Maneuvering, 4000 lb.)	146 m.p.h.	(127 knots)
V_A (Maneuvering, 2743 lb.)	133 m.p.h	(115 knots)
V_{FE} (Flaps extended)	125 m.p.h	(109 knots)
V_{LO} (Landing gear operating)		
Extension	150 m.p.h.	(130 knots)
Retract	125 m.p.h.	(109 knots)
V_{LE} (Landing gear extended)	150 m.p.h	(130 knots)
V_{MC} (Minimum control speed)	80 m.p.h.	(69 knots)

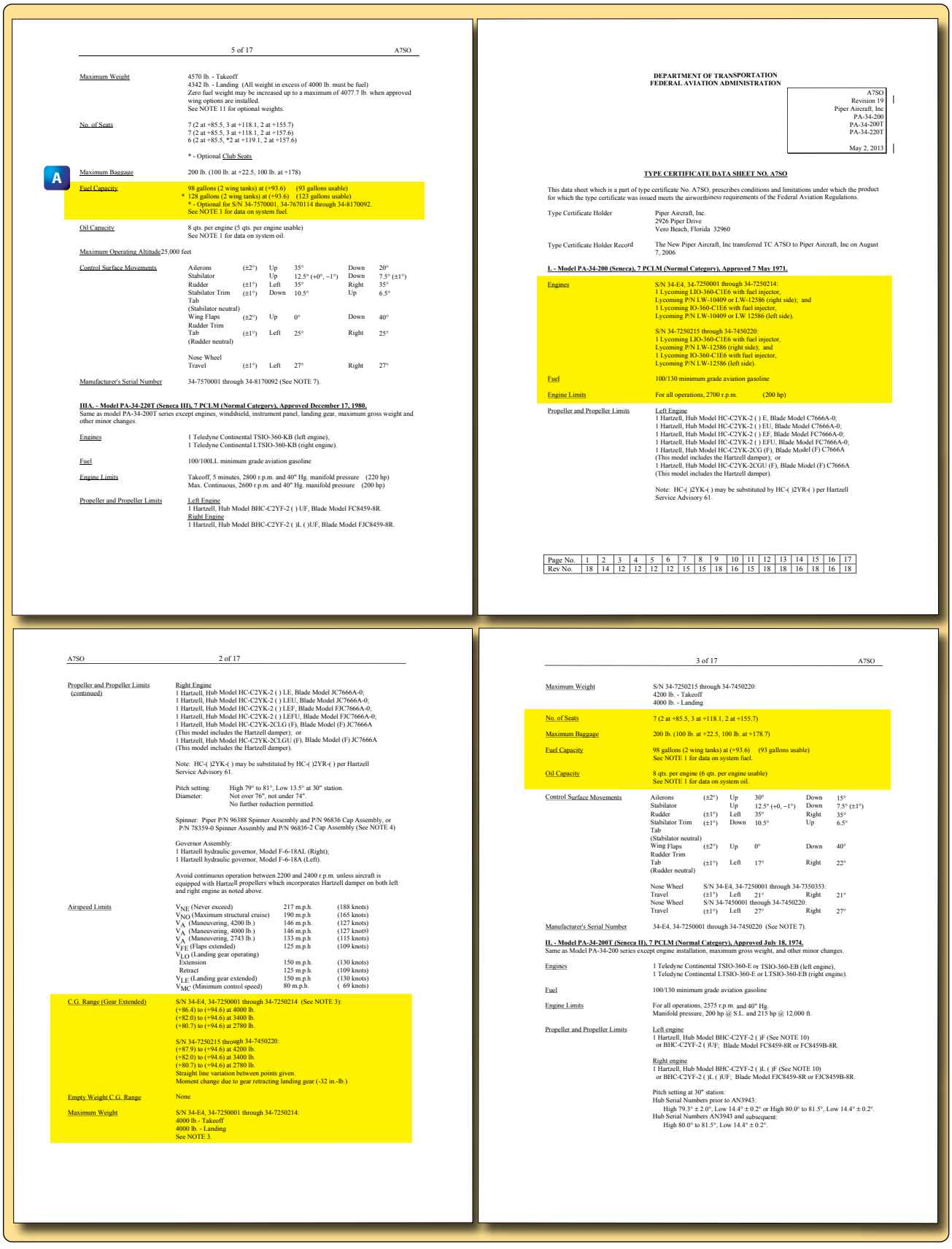
C.G. Range (Gear Extended) — S/N 34-E4, 34-7250001 through 34-7250214 (See NOTE 3):
(+86.4) to (+94.6) at 4000 lb.
(+82.0) to (+94.6) at 3400 lb.
(+80.7) to (+94.6) at 2780 lb.

S/N 34-7250215 through 34-7450220:
(+87.9) to (+94.6) at 4200 lb.
(+82.0) to (+94.6) at 3400 lb.
(+80.7) to (+94.6) at 2780 lb.
Straight line variation between points given.
Moment change due to gear retracting landing gear (-32 in.-lb.)

Empty Weight C.G. Range — None

Maximum Weight — S/N 34-E4, 34-7250001 through 34-7250214:
4000 lb.- Takeoff
4000 lb. - Landing
See NOTE 3.

Figure 6-14. *The Type Certificate Data Sheet (TCDS) shows various information about an aircraft to include weight and balance information.*



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Maximum Weight
4570 lb. - Takeoff
4342 lb. - Landing (All weight in excess of 4000 lb. must be fuel)
Zero fuel weight may be increased up to a maximum of 4077.7 lb. when approved wing options are installed.
See NOTE 11 for optional weights.

No. of Seats
7 (2 at +85.5, 3 at +118.1, 2 at +155.7)
7 (2 at +85.5, 3 at +118.1, 2 at +157.6)
6 (2 at +85.5, *2 at +119.1, 2 at +157.6)

* - Optional Club Seats

A

Maximum Baggage
200 lb. (100 lb. at +22.5, 100 lb. at +178)

Fuel Capacity
98 gallons (2 wing tanks) at (+93.6) (93 gallons usable)
* 128 gallons (2 wing tanks) at (+93.6) (123 gallons usable)
* - Optional for S/N 34-7570001, 34-7670114 through 34-8170092.
See NOTE 1 for data on system fuel.

Oil Capacity
8 qts. per engine (5 qts. per engine usable)
See NOTE 1 for data on system oil.

Maximum Operating Altitude 25,000 feet

Control Surface Movements						
Ailerons	(±2°)	Up	35°	Down	20°
Stabilator		Up	12.5° (+0°, −1°)	Down	7.5° (±1°)
Rudder	(±1°)	Left	35°	Right	35°
Stabilator Trim Tab (Stabilator neutral)	(±1°)	Down	10.5°	Up	6.5°
Wing Flaps	(±2°)	Up	0°	Down	40°
Rudder Trim Tab (Rudder neutral)	(±1°)	Left	25°	Right	25°
Nose Wheel Travel	(±1°)	Left	27°	Right	27°

Manufacturer's Serial Number
34-7570001 through 34-8170092 (See NOTE 7).

IIIA. - Model PA-34-220T (Seneca III), 7 PCLM (Normal Category), Approved December 17, 1980.
Same as model PA-34-200T series except engines, windshield, instrument panel, landing gear, maximum gross weight and other minor changes.

Engines
1 Teledyne Continental TSIO-360-KB (left engine),
1 Teledyne Continental LTSIO-360-KB (right engine).

Fuel
100/100LL minimum grade aviation gasoline

Engine Limits
Takeoff, 5 minutes, 2800 r.p.m. and 40" Hg. manifold pressure (220 hp)
Max. Continuous, 2600 r.p.m. and 40" Hg. manifold pressure (200 hp)

Propeller and Propeller Limits
Left Engine
1 Hartzell, Hub Model BHC-C2YF-2 ( ) UF, Blade Model FC8459-8R.
Right Engine
1 Hartzell, Hub Model BHC-C2YF-2 ( )L ( )UF, Blade Model FJC8459-8R.

DEPARTMENT OF TRANSPORTATION
FEDERAL AVIATION ADMINISTRATION

A7SO
Revision 19
Piper Aircraft, Inc
PA-34-200
PA-34-200T
PA-34-220T

May 2, 2013

TYPE CERTIFICATE DATA SHEET NO. A7SO

This data sheet which is a part of type certificate No. A7SO, prescribes conditions and limitations under which the product for which the type certificate was issued meets the airworthiness requirements of the Federal Aviation Regulations.

Type Certificate Holder
Piper Aircraft, Inc.
2926 Piper Drive
Vero Beach, Florida 32960

Type Certificate Holder Record
The New Piper Aircraft, Inc transferred TC A7SO to Piper Aircraft, Inc on August 7, 2006

I. - Model PA-34-200 (Seneca), 7 PCLM (Normal Category), Approved 7 May 1971.

Engines
S/N 34-E4, 34-7250001 through 34-7250214:
1 Lycoming LIO-360-C1E6 with fuel injector,
Lycoming P/N LW-10409 or LW-12586 (right side); and
1 Lycoming IO-360-C1E6 with fuel injector,
Lycoming P/N LW-10409 or LW 12586 (left side).

S/N 34-7250215 through 34-7450220:
1 Lycoming LIO-360-C1E6 with fuel injector,
Lycoming P/N LW-12586 (right side); and
1 Lycoming IO-360-C1E6 with fuel injector,
Lycoming P/N LW-12586 (left side).

Fuel
100/130 minimum grade aviation gasoline

Engine Limits
For all operations, 2700 r.p.m. (200 hp)

Propeller and Propeller Limits
Left Engine
1 Hartzell, Hub Model HC-C2YK-2 ( ) E, Blade Model C7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) EU, Blade Model C7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) EF, Blade Model FC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) EFU, Blade Model FC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2CG (F), Blade Model (F) C7666A
(This model includes the Hartzell damper); or
1 Hartzell, Hub Model HC-C2YK-2CGU (F), Blade Model (F) C7666A
(This model includes the Hartzell damper).

Note: HC-( )2YK-( ) may be substituted by HC-( )2YR-( ) per Hartzell Service Advisory 61.

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Propeller and Propeller Limits (continued)
Right Engine
1 Hartzell, Hub Model HC-C2YK-2 ( ) LE, Blade Model JC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) LEU, Blade Model JC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) LEF, Blade Model FJC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2 ( ) LEFU, Blade Model FJC7666A-0;
1 Hartzell, Hub Model HC-C2YK-2CLG (F), Blade Model (F) JC7666A
(This model includes the Hartzell damper); or
1 Hartzell, Hub Model HC-C2YK-2CLGU (F), Blade Model (F) JC7666A
(This model includes the Hartzell damper).

Note: HC-( )2YK-( ) may be substituted by HC-( )2YR-( ) per Hartzell Service Advisory 61.

Pitch setting: High 79° to 81°, Low 13.5° at 30" station.
Diameter: Not over 76", not under 74".
No further reduction permitted.

Spinner: Piper P/N 96388 Spinner Assembly and P/N 96836 Cap Assembly, or P/N 78359-0 Spinner Assembly and P/N 96836-2 Cap Assembly (See NOTE 4)

Governor Assembly:
1 Hartzell hydraulic governor, Model F-6-18AL (Right);
1 Hartzell hydraulic governor, Model F-6-18A (Left).

Avoid continuous operation between 2200 and 2400 r.p.m. unless aircraft is equipped with Hartzell propellers which incorporates Hartzell damper on both left and right engine as noted above.

Airspeed Limits

V_{NE} (Never exceed)	217 m.p.h.	(188 knots)
V_{NO} (Maximum structural cruise)	190 m.p.h	(165 knots)
V_A (Maneuvering, 4200 lb.)	146 m.p.h.	(127 knots)
V_A (Maneuvering, 4000 lb.)	146 m.p.h.	(127 knots)
V_A (Maneuvering, 2743 lb.)	133 m.p.h	(115 knots)
V_{FE} (Flaps extended)	125 m.p.h	(109 knots)
V_{LO} (Landing gear operating)		
Extension	150 m.p.h.	(130 knots)
Retract	125 m.p.h	(109 knots)
V_{LE} (Landing gear extended)	150 m.p.h	(130 knots)
V_{MC} (Minimum control speed)	80 m.p.h.	(69 knots)

C.G. Range (Gear Extended)
S/N 34-E4, 34-7250001 through 34-7250214 (See NOTE 3):
(+86.4) to (+94.6) at 4000 lb.
(+82.0) to (+94.6) at 3400 lb.
(+80.7) to (+94.6) at 2780 lb.

S/N 34-7250215 through 34-7450220:
(+87.9) to (+94.6) at 4200 lb.
(+82.0) to (+94.6) at 3400 lb.
(+80.7) to (+94.6) at 2780 lb.
Straight line variation between points given.
Moment change due to gear retracting landing gear (-32 in.-lb.)

Empty Weight C.G. Range
None

Maximum Weight
S/N 34-E4, 34-7250001 through 34-7250214:
4000 lb.- Takeoff
4000 lb. - Landing
See NOTE 3.

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Maximum Weight
S/N 34-7250215 through 34-7450220:
4200 lb. - Takeoff
4000 lb. - Landing

No. of Seats
7 (2 at +85.5, 3 at +118.1, 2 at +155.7)

Maximum Baggage
200 lb. (100 lb. at +22.5, 100 lb. at +178.7)

Fuel Capacity
98 gallons (2 wing tanks) at (+93.6) (93 gallons usable)
See NOTE 1 for data on system fuel.

Oil Capacity
8 qts. per engine (6 qts. per engine usable)
See NOTE 1 for data on system oil.

Control Surface Movements						
Ailerons	(±2°)	Up	30°	Down	15°
Stabilator		Up	12.5° (+0, −1°)	Down	7.5° (±1°)
Rudder	(±1°)	Left	35°	Right	35°
Stabilator Trim Tab (Stabilator neutral)	(±1°)	Down	10.5°	Up	6.5°
Wing Flaps	(±2°)	Up	0°	Down	40°
Rudder Trim Tab (Rudder neutral)	(±1°)	Left	17°	Right	22°
Nose Wheel Travel	S/N 34-E4, 34-7250001 through 34-7350353: (±1°)	Left	21°	Right	21°
Nose Wheel Travel	S/N 34-7450001 through 34-7450220: (±1°)	Left	27°	Right	27°

Manufacturer's Serial Number
34-E4, 34-7250001 through 34-7450220 (See NOTE 7).

II. - Model PA-34-200T (Seneca II), 7 PCLM (Normal Category), Approved July 18, 1974.
Same as Model PA-34-200 series except engine installation, maximum gross weight, and other minor changes.

Engines
1 Teledyne Continental TSIO-360-E or TSIO-360-EB (left engine),
1 Teledyne Continental LTSIO-360-E or LTSIO-360-EB (right engine).

Fuel
100/130 minimum grade aviation gasoline

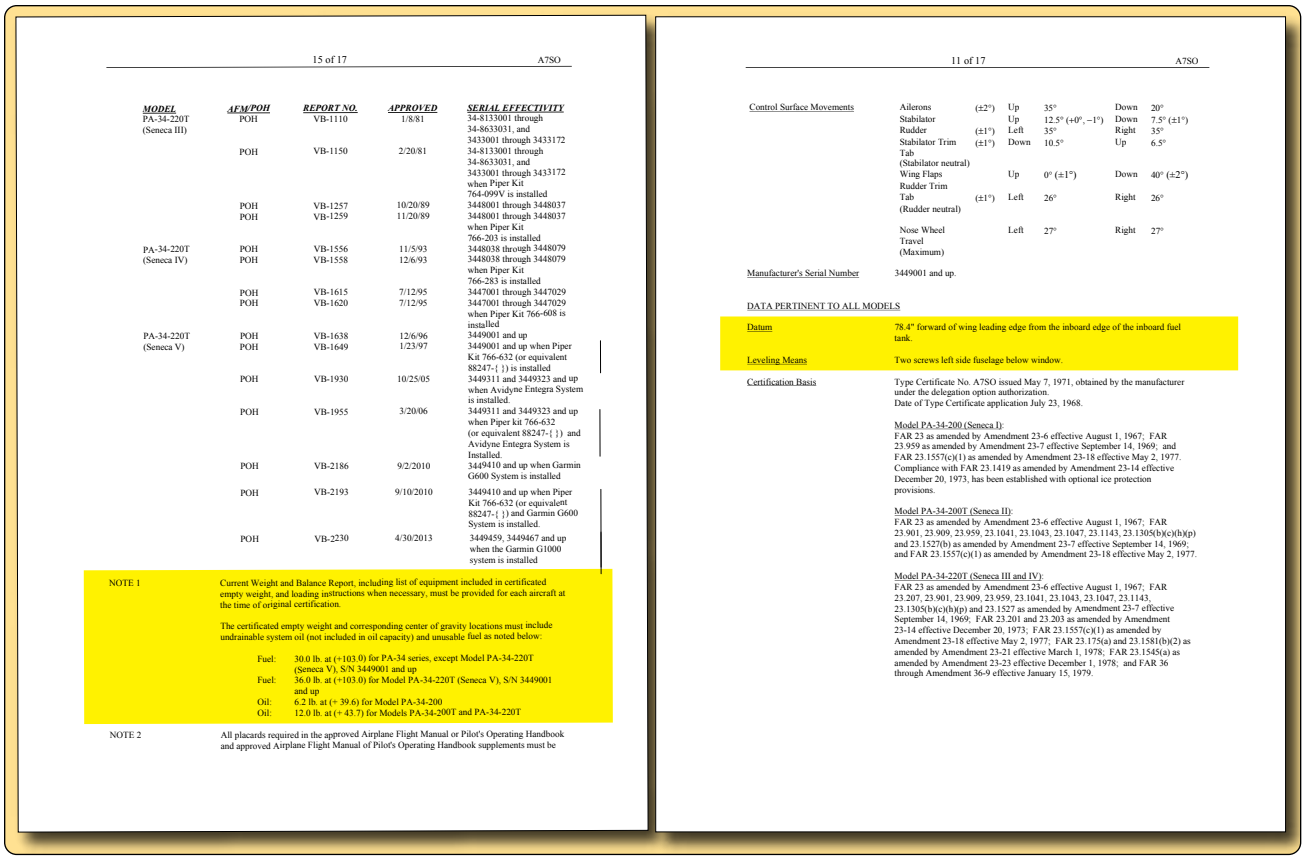
Engine Limits
For all operations, 2575 r.p.m. and 40" Hg.
Manifold pressure, 200 hp @ S.L. and 215 hp @ 12,000 ft.

Propeller and Propeller Limits
Left engine
1 Hartzell, Hub Model BHC-C2YF-2 ( )F (See NOTE 10)
or BHC-C2YF-2 ( )UF; Blade Model FC8459-8R or FC8459B-8R.

Right engine
1 Hartzell, Hub Model BHC-C2YF-2 ( )L ( )F (See NOTE 10)
or BHC-C2YF-2 ( )L ( )UF; Blade Model FJC8459-8R or FJC8459B-8R.

Pitch setting at 30" station:
Hub Serial Numbers prior to AN3943:
High 79.3° ± 2.0°, Low 14.4° ± 0.2° or High 80.0° to 81.5°, Low 14.4° ± 0.2°.
Hub Serial Numbers AN3943 and subsequent:
High 80.0° to 81.5°, Low 14.4° ± 0.2°.

Figure 6-15. *Highlights of various specifications of the aircraft found within a TCDS. Note the fuel capacity of 98 gallons and its reference to the datum (A).*



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MODEL	AFM/POH	REPORT NO.	APPROVED	SERIAL EFFECTIVITY
PA-34-220T (Seneca III)	POH	VB-1110	1/8/81	34-8133001 through 34-8633031, and 3433001 through 3433172
	POH	VB-1150	2/20/81	34-8133001 through 34-8633031, and 3433001 through 3433172 when Piper Kit 764-099V is installed
	POH	VB-1257	10/20/89	3448001 through 3448037
	POH	VB-1259	11/20/89	3448001 through 3448037 when Piper Kit 766-203 is installed
PA-34-220T (Seneca IV)	POH	VB-1556	11/5/93	3448038 through 3448079
	POH	VB-1558	12/6/93	3448038 through 3448079 when Piper Kit 766-283 is installed
	POH	VB-1615	7/12/95	3447001 through 3447029
	POH	VB-1620	7/12/95	3447001 through 3447029 when Piper Kit 766-608 is installed
PA-34-220T (Seneca V)	POH	VB-1638	12/6/96	3449001 and up
	POH	VB-1649	1/23/97	3449001 and up when Piper Kit 766-632 (or equivalent 88247-[ ]) is installed
	POH	VB-1930	10/25/05	3449311 and 3449323 and up when Avidyne Entegra System is installed.
	POH	VB-1955	3/20/06	3449311 and 3449323 and up when Piper kit 766-632 (or equivalent 88247-[ ]) and Avidyne Entegra System is Installed.
	POH	VB-2186	9/2/2010	3449410 and up when Garmin G600 System is installed
	POH	VB-2193	9/10/2010	3449410 and up when Piper Kit 766-632 (or equivalent 88247-[ ]) and Garmin G600 System is installed.
	POH	VB-2230	4/30/2013	3449459, 3449467 and up when the Garmin G1000 system is installed

NOTE 1 Current Weight and Balance Report, including list of equipment included in certificated empty weight, and loading instructions when necessary, must be provided for each aircraft at the time of original certification.

The certificated empty weight and corresponding center of gravity locations must include undrainable system oil (not included in oil capacity) and unusable fuel as noted below:

Fuel: 30.0 lb. at (+103.0) for PA-34 series, except Model PA-34-220T (Seneca V), S/N 3449001 and up
Fuel: 36.0 lb. at (+103.0) for Model PA-34-220T (Seneca V), S/N 3449001 and up
Oil: 6.2 lb. at (+ 39.6) for Model PA-34-200
Oil: 12.0 lb. at (+ 43.7) for Models PA-34-200T and PA-34-220T

NOTE 2 All placards required in the approved Airplane Flight Manual or Pilot's Operating Handbook and approved Airplane Flight Manual of Pilot's Operating Handbook supplements must be

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Control Surface Movements	Ailerons	(±2°)	Up	35°	Down	20°
	Stabilator		Up	12.5° (+0°, –1°)	Down	7.5° (±1°)
	Rudder	(±1°)	Left	35°	Right	35°
	Stabilator Trim Tab (Stabilator neutral)	(±1°)	Down	10.5°	Up	6.5°
	Wing Flaps		Up	0° (±1°)	Down	40° (±2°)
	Rudder Trim Tab (Rudder neutral)	(±1°)	Left	26°	Right	26°
	Nose Wheel Travel (Maximum)		Left	27°	Right	27°

Manufacturer's Serial Number 3449001 and up.

DATA PERTINENT TO ALL MODELS

Datum 78.4" forward of wing leading edge from the inboard edge of the inboard fuel tank.

Leveling Means Two screws left side fuselage below window.

Certification Basis Type Certificate No. A7SO issued May 7, 1971, obtained by the manufacturer under the delegation option authorization.
Date of Type Certificate application July 23, 1968.

Model PA-34-200 (Seneca I):
FAR 23 as amended by Amendment 23-6 effective August 1, 1967; FAR 23.959 as amended by Amendment 23-7 effective September 14, 1969; and FAR 23.1557(c)(1) as amended by Amendment 23-18 effective May 2, 1977. Compliance with FAR 23.1419 as amended by Amendment 23-14 effective December 20, 1973, has been established with optional ice protection provisions.

Model PA-34-200T (Seneca II):
FAR 23 as amended by Amendment 23-6 effective August 1, 1967; FAR 23.901, 23.909, 23.959, 23.1041, 23.1043, 23.1047, 23.1143, 23.1305(b)(c)(h)(p) and 23.1527(b) as amended by Amendment 23-7 effective September 14, 1969; and FAR 23.1557(c)(1) as amended by Amendment 23-18 effective May 2, 1977.

Model PA-34-220T (Seneca III and IV):
FAR 23 as amended by Amendment 23-6 effective August 1, 1967; FAR 23.207, 23.901, 23.909, 23.959, 23.1041, 23.1043, 23.1047, 23.1143, 23.1305(b)(c)(h)(p) and 23.1527 as amended by Amendment 23-7 effective September 14, 1969; FAR 23.201 and 23.203 as amended by Amendment 23-14 effective December 20, 1973; FAR 23.1557(c)(1) as amended by Amendment 23-18 effective May 2, 1977; FAR 23.175(a) and 23.1581(b)(2) as amended by Amendment 23-21 effective March 1, 1978; FAR 23.1545(a) as amended by Amendment 23-23 effective December 1, 1978; and FAR 36 through Amendment 36-9 effective January 15, 1979.

Figure 6-16. *Highlights of various specifications of the aircraft found within a TCDS.*

is weighed using load cells on jacks, leveling the aircraft is done by adjusting the height with the jacks and checking the level at the level point. *Figure 6-20* shows a Gulfstream jet on jacks with the load cells in place.

Always follow the aircraft manufacturer's weighing and leveling procedures and processes. All aircraft need to be in a flight level attitude when they are weighed unless the manufacturer's manual specifically allows it or has a formula in the manual to use accordingly.

Spirit Level

Before an aircraft can be weighed and reliable readings obtained, it must be in a level flight attitude. One method that can be used to check for a level condition is to use a spirit level, sometimes thought of as a carpenter's level, by placing it on or against a specified place on the aircraft. Spirit levels consist of a vial full of liquid, except for a small air bubble. When the air bubble is centered between the two black lines, a level condition is indicated.

In *Figure 6-21,* a spirit level is being used on a Mooney M20 to check for a flight level attitude. By looking in the TCDS, it is determined that the leveling means is two screws on the left side of the airplane fuselage, in line with the trailing edge of the wing.

Plumb Bob

A plumb bob is a heavy metal object, cylinder or cone shape, with a sharp point at one end and a string attached to the other end. If the string is attached to a given point on an aircraft, and the plumb bob can hang down so the tip just touches the ground, the point where the tip touches will be perpendicular to where the string is attached. An example of the use of a plumb bob would be measuring the distance from an aircraft's datum to the center of the main landing gear axle. If the leading edge of the wing was the datum, a plumb bob could be dropped from the leading edge and a



Figure 6-17. *Weighing a Piper Archer using electronic platform scales.*



Figure 6-18. *A Cessna 182 being weighed with portable electronic platform scales.*

chalk mark made on the hangar floor. The plumb bob could also be dropped from the center of the axle on the main landing gear, and a chalk mark made on the floor. With a tape measure, the distance between the two chalk marks could be determined, and the arm for the main landing gear would be known. Plumb bobs can also be used to level an aircraft, as described in the Helicopter Weight and Balance section of this chapter. *Figure 6-22* shows a plumb bob being dropped from the leading edge of an aircraft wing.

Hydrometer

When an aircraft is weighed with full fuel in the tanks, the weight of the fuel must be accounted for by mathematically subtracting it from the scale readings. To subtract it, its weight, arm, and moment must be known. Although the standard weight for aviation gasoline (Avgas) is 6.0 lb/gal and jet fuel is 6.7 lb/gal, these values are not exact for all conditions. On a hot day versus a cold day, these values can vary dramatically. On a hot summer day in the state of Florida, Avgas checked with a hydrometer typically weighs between 5.85 and 5.9 lb/gal. If 100 gallons of fuel were involved in a calculation, using the actual weight versus the standard weight would make a difference of 10 to 15 lb.



Figure 6-19. *M2000 platform scale digital indicator.*

When an aircraft is weighed with fuel in the tanks, the weight of fuel per gallon should be checked with a hydrometer. A hydrometer consists of a weighted glass tube that is sealed with a graduated set of markings on the side of the tube. The graduated markings and their corresponding number values represent units of pounds per gallon (lb/gal). When placed in a flask with fuel in it, the glass tube floats at a level dependent on the density of the fuel. Where the fuel intersects the markings on the side of the tube indicates the pounds per gallon.

Preparing an Aircraft for Weighing

Weighing an aircraft is a very important and exacting phase of aircraft maintenance and must be carried out with accuracy and good workmanship. Thoughtful preparation saves time and prevents mistakes. The aircraft should be weighed inside a hangar where wind cannot blow over the surface and cause fluctuating or false scale readings. The aircraft should be clean inside and out, with special attention paid to the bilge area to be sure no water or debris is trapped. The outside of the aircraft should be as free as possible of all mud and dirt.

To begin, assemble all the necessary equipment, such as:

1. Scales, hoisting equipment, jacks, and leveling equipment.
2. Blocks, chocks, or sandbags for holding the airplane



Figure 6-20. *Airplane on jacks with load cells in use.*

on the scales.

3. Straightedge, spirit level, plumb bobs, chalk line, and a measuring tape.
4. Applicable Aircraft Specifications and weight and balance computation forms.

Fuel System

When weighing an aircraft to determine its empty weight, only the weight of residual (unusable) fuel should be included. To ensure that only residual fuel is accounted for, the aircraft should be weighed in one of the following three conditions.

1. Weigh the aircraft with absolutely no fuel in the aircraft tanks or fuel lines. If an aircraft is weighed in this condition, the technician can mathematically add the proper amount of residual fuel to the aircraft and account for its arm and moment. The proper amount of fuel can be determined by looking in the aircraft's TCDS.
2. Drain fuel from the tanks in the manner specified by the aircraft manufacturer. If there are no specific instructions, drain the fuel until the fuel quantity gauges read empty and until fuel stops draining from the tanks. The aircraft attitude may be a consideration when draining the fuel tanks and the maintenance manual should be consulted. In this case, the unusable fuel will remain in the lines and system, and its weight and arm can be determined by reference to the aircraft's TCDS.
3. Weigh the aircraft with the fuel tanks completely full. If an aircraft is weighed in this condition, the technician can mathematically subtract the weight of usable fuel and account for its arm and moment. If the weight of the fuel is in question, a hydrometer can also be used to determine the weight of each gallon of fuel, while the Aircraft Specifications or TCDS can be used to identify the fuel capacity of the aircraft. If an aircraft is to be weighed with load cells attached to jacks, the technician should check both the load cell instruction manual and aircraft maintenance manual to make sure it is permissible to jack the aircraft with the fuel tanks full as this may add additional stress to the aircraft structure.



Figure 6-21. *Spirit level being used on a Mooney M20.*



Figure 6-22. *Plumb bob dropped from a wing leading edge.*

Never weigh an aircraft with fuel tanks partially full, because it will be impossible to determine exactly how much fuel to account for.

Oil System

The empty weight for older aircraft certificated under the Civil Air Regulations (CAR) part 3 does not include the engine lubricating oil. The oil must be drained before the aircraft is weighed, or its weight must be subtracted from the scale readings to determine the empty weight.

To weigh an aircraft that does not include the engine lubricating oil as part of the empty weight, place it in level flight attitude, then open the drain valves and allow all the oil that is able, to drain out. Any remaining is undrainable oil and is part of the empty weight.

If it is impractical to drain the oil, the reservoir can be filled to the specified level and the weight of the oil computed at 7.5 lb/gal. Then its weight and moment are subtracted from the weight and moment of the aircraft as weighed. The amount and arm of the undrainable oil are found in NOTE 1 of the TCDS, and this must be added to the empty weight.

For aircraft certificated since 1978 under 14 CFR parts 23 and 25, full engine oil is typically included in an aircraft's

empty weight. This can be confirmed by looking at the TCDS. If full oil is to be included, the oil level needs to be checked and the oil system serviced if it is less than full.

Miscellaneous Fluids

The hydraulic fluid reservoir and all other reservoirs containing fluids required for normal operation of the aircraft should be full. Fluids not considered to be part of the empty weight of the aircraft are potable (drinkable) water, lavatory precharge water, and water for injection into the engines.

Flight Controls

The position of such items as spoilers, slats, flaps, and helicopter rotor systems is an important factor when weighing an aircraft. Always refer to the manufacturer's instructions for the proper position of these items.

Other Considerations

Inspect the aircraft to see that all items included in the certificated empty weight are installed in the proper location. Remove items that are not regularly carried in flight. Also, look in the baggage compartments to make sure they are empty. Replace all inspection plates, oil and fuel tank caps, junction box covers, cowling, doors, emergency exits, and other parts that have been removed during maintenance. All doors, windows, and sliding canopies should be in the normal flight position. Remove excessive dirt, oil, grease, and moisture from the aircraft.

Some aircraft are not weighed with the wheels on the scales, but are weighed with the scales placed either at the jacking points or at special weighing points. Regardless of what provisions are made for placing the aircraft on the scales or jacks, be careful to prevent it from falling or rolling off, thereby damaging the aircraft and equipment. When weighing an aircraft with the wheels placed on the scales, release the brakes to reduce the possibility of incorrect readings caused by side loads on the scales.

All aircraft have leveling points or lugs, and care must be taken to level the aircraft, especially along the longitudinal axis. With light, fixed-wing airplanes, the lateral level is not as critical as it is with heavier airplanes. However, a reasonable effort should be made to level the light airplanes along the lateral axis. Helicopters must be level longitudinally and laterally when they are weighed. Accuracy in leveling all aircraft longitudinally cannot be overemphasized.

Weighing Points

When an aircraft is being weighed, the arms must be known for the points where the weight of the aircraft is being transferred to the scales. If a tricycle gear small airplane has its three wheels sitting on floor scales, the weight transfer to each scale happens through the center of the axle for each wheel. If an airplane is weighed while it is on jacks, the weight transfer happens through the center of the jack pad. For a helicopter with skids for landing gear, determining the arm for the weighing points can be difficult if the skids are sitting directly on floor scales. The problem is that the skid is in contact with the entire top portion of the scale, and it is impossible to know exactly where the center of weight transfer is occurring. In such a case, place a piece of pipe between the skid and the scale, and the center of the pipe will now be the known point of weight transfer.

The arm for each of the weighing points is the distance from the center of the weight transfer point to the aircraft's datum. If the arms are not known, based on previous weighing of the aircraft or some other source of data, they must be measured when the aircraft is weighed. This involves dropping a plumb bob from the center of each weighing point and from the aircraft datum, and putting a chalk mark on the hangar floor representing each point. The perpendicular distance between the datum and each of the weighing points can then be measured. In *Figure 6-23,* the distance from the nosewheel centerline to the datum is being measured on an airplane. The nosewheel sitting on an electronic scale can be seen in the background.

Jacking the Aircraft

Aircraft are often weighed by rolling them onto ramps in which load cells are embedded. This eliminates the problems associated with jacking the aircraft off the ground. However, many aircraft are weighed by jacking the aircraft up and then lowering them onto scales or load cells. Extra care must be used when raising an aircraft on jacks for weighing. If the aircraft has spring steel landing gear and it is jacked at the wheel, the landing gear will slide inward as the weight is taken off the tire. Care must be taken to prevent the jack from tipping over. For some aircraft, stress panels or plates must be installed before they are raised with wing jacks to distribute the weight over the jack pad. Be sure to follow the recommendations of the aircraft manufacturer in detail anytime an aircraft is jacked. When using two wing jacks, take special care to raise them simultaneously, so the aircraft does not slip off the jacks. As the jacks are raised, keep the safety collars screwed down against the jack cylinder to prevent the aircraft from tilting if one of the jacks should lose hydraulic pressure.

Leveling the Aircraft

When an aircraft is weighed, it must be in its level flight attitude so that all the components are at the correct distance from the datum. This attitude is determined by information in the TCDS. Some aircraft require a plumb line to be dropped

from a specified location so that the point of the weight, the bob, hangs directly above an identifiable point. Others specify that a spirit level be placed across two leveling lugs (special screws on the outside of the fuselage). Other aircraft call for a spirit level to be placed on the upper door sill. Lateral level is not specified for all light aircraft, but provisions are normally made on helicopters for determining both longitudinal and lateral level. This may be done by built-in leveling indicators or by a plumb bob that shows the conditions of both longitudinal and lateral level. The actual adjustments to level the aircraft using load cells are made with the jacks. When weighing from the wheels, leveling is normally done by adjusting the air pressure in the nosewheel shock strut.

Safety Considerations

Special precautions must be taken when raising an aircraft on jacks.

1. Stress plates must be installed under the jack pads if the manufacturer specifies them.
2. If anyone is required to be in the aircraft while it is being jacked, there must be no movement.
3. The jacks must be straight under the jack pads before beginning to raise the aircraft.
4. All jacks must be raised simultaneously and safety devices placed against the jack cylinder to prevent the aircraft from tipping if any jack should lose pressure. Not all jacks have screw-down collars, some use drop pins or friction locks.

CG Range

The CG range for an aircraft is the limits within which the aircraft must balance. It is identified as a range and considered an arm extending from the forward most limit to the aft most limit usually expressed in inches. In the TCDS for the Piper Seneca airplane, shown earlier in this chapter, the range is given in *Figure 6-24.*

Because the Piper Seneca is a retractable gear airplane, the specifications identify that the range applies when the landing gear is extended, and that the airplane's total moment is decreased by 32 when the gear retracts. To know how much the CG changes when the gear is retracted, the moment of 32 in-lb would need to be divided by the loaded weight of the airplane. For example, if the airplane weighed 3,500 lb, the CG would move forward 0.009" (32 ÷ 3,500).

Based on the numbers given, up to a loaded weight of 2,780 lb, the forward CG limit is +80.7" and the aft CG limit is +94.6". As the loaded weight of the airplane increases to 3,400 lb, and eventually to the maximum of 4,000 lb, the forward CG limit moves aft. In other words, as the loaded weight of the airplane increases, the CG range gets smaller. The range gets smaller because of the forward limit moving back, while the aft limit stays in the same place.

The data sheet identifies that there is a straight-line variation between the points given. The points being referred to are the forward and aft CG limits. From a weight of 2,780 lb to a weight of 3,400 lb, the forward limit moves from +80.7" to +82.0", and if plotted on a graph, that change would form a straight line. From 3,400 lb to 4,000 lb, the forward limit moves from +82 to +86.4", again forming a straight line. Plotted on a graph, the CG limits would look like *Figure 6-25.* When graphically plotted, the CG limits form what is known as the CG envelope.

In *Figure 6-25,* the red line represents the forward limit up to a weight of 2,780 lb. The blue and green lines represent the straight-line variation that occurs for the forward limit as the weight increases up to a maximum of 4,000 lb. The yellow line represents the maximum weight for the airplane, and the purple line represents the aft limit.

Empty Weight Center of Gravity (EWCG) Range

For some aircraft, a CG range is given for the aircraft in the empty weight condition in the TCDS. This practice is not very common with airplanes, but is often done for helicopters. This range would only be listed for an airplane if the fuel tanks, seats, and baggage compartments are so located that changes in the fuel or occupant load have a very limited effect on the balance of the aircraft. If the EWCG of an aircraft falls within the EWCG limits, it is impossible to legally load the aircraft so that its loaded CG falls outside of its allowable range. If the TCDS lists an EWCG range and, after a repair or alteration is completed, the EWCG falls within this range, then there is no need to compute a fore and aft check for adverse loading. But if the TCDS lists the EWCG range as "None" (and most of them do), a check must be made to determine whether it is possible by any combination of legal loading to cause the



Figure 6-23. *Measuring the nosewheel arm on an airplane.*

aircraft CG to move outside of either its forward or aft limits.

Operating CG Range

All aircraft have CG limits identified for the operational condition, with the aircraft loaded and ready for flight. If an aircraft can operate in more than one category, such as normal and utility, more than one set of limits might be listed. As shown earlier for the Piper Seneca airplane, the limits can change as the weight of the aircraft increases. To legally fly, the CG for the aircraft must fall within the CG limits.

Standard Weights Used for Aircraft Weight and Balance

Unless the specific weight for an item is known, the standard weights used in aircraft weight and balance are as follows:

- Avgas 6 lb/gal
- Turbine fuel 6.7 lb/gal
- Lubricating oil 7.5 lb/gal
- Water 8.35 lb/gal
- Crew and passengers 170 lb per person

Example Weighing of an Airplane

In *Figure 6-26,* a tricycle gear airplane is being weighed by using three floor scales. The specifications on the airplane and the weighing specific data are shown in *Figure 6-27.*

By analyzing the data identified for the airplane being weighed in *Figure 6-26,* the following information is determined.

- Because the airplane was weighed with the fuel tanks full, the full weight of the fuel must be subtracted and the unusable fuel added back in. The weight of the fuel being subtracted is based on the pounds per gallon determined by the hydrometer check (5.9 lb/gal).
- Because wheel chocks are used to keep the airplane from rolling off the scales, their weight must be subtracted from the scale readings as tare weight.
- Because the main wheel centerline is 70" behind the datum, its arm is a +70".
- The arm for the nosewheel is the difference between the wheelbase (100") and the distance from the datum to the main wheel centerline (70"). Therefore, the arm for the nosewheel is −30".

To calculate the airplane's empty weight and EWCG, a six-column chart is used. *Figure 6-28* shows the calculation for the airplane in *Figure 6-26.*

Based on the calculation shown in the chart, the CG is at +50.1", which means it is 50.1" aft of the datum. This places the CG forward of the main landing gear, which must be the case for a tricycle gear airplane. This number is the result of dividing the total moment of 66,698 in-lb by the total weight of 1,331.5 lb.

CG Range: (Gear Extended)
S/N 34-E4, 34-7250001 through 34-7250214
(See NOTE 3)
(+86.4") to (+94.6") at 4,000 lb
(+82.0") to (+94.6") at 3,400 lb
(+80.7") to (+94.6") at 2,780 lb
Straight line variation between points given.
Moment change due to gear retracting landing gear (–32 in-lb)

Figure 6-24. *Piper Seneca airplane center of gravity range.*

EWCG Formulas

The EWCG can be quickly calculated by using the following formulas. There are four possible conditions and formulas that relate the location of the CG to the datum. Notice that the formula for each condition first determines the moment of the nose $\frac{F \times L}{W}$ wheel or tail $\frac{R \times L}{W}$ wheel and then divides it by the total weight of the airplane. The arm is then added to or subtracted from the distance between the main wheels and the datum (distance D).

Formula 1 Nosewheel airplanes with datum forward of the main wheels.

$$CG = D - \left(\frac{F \times L}{W}\right)$$

Formula 2 Nosewheel airplanes with the datum aft of the main wheels.

$$CG = -\left(D + \frac{F \times L}{W}\right)$$

Formula 3 Tail wheel airplanes with the datum forward of the main wheels.

$$CG = D + \left(\frac{R \times L}{W}\right)$$

Formula 4 Tail wheel airplanes with the datum aft of the main wheels.

$$CG = -D + \left(\frac{R \times L}{W}\right)$$

Datum Forward of the Airplane–Nosewheel Landing Gear

The datum of the airplane in *Figure 6-29* is 100" forward of the leading edge of the wing root, or 128" forward of the

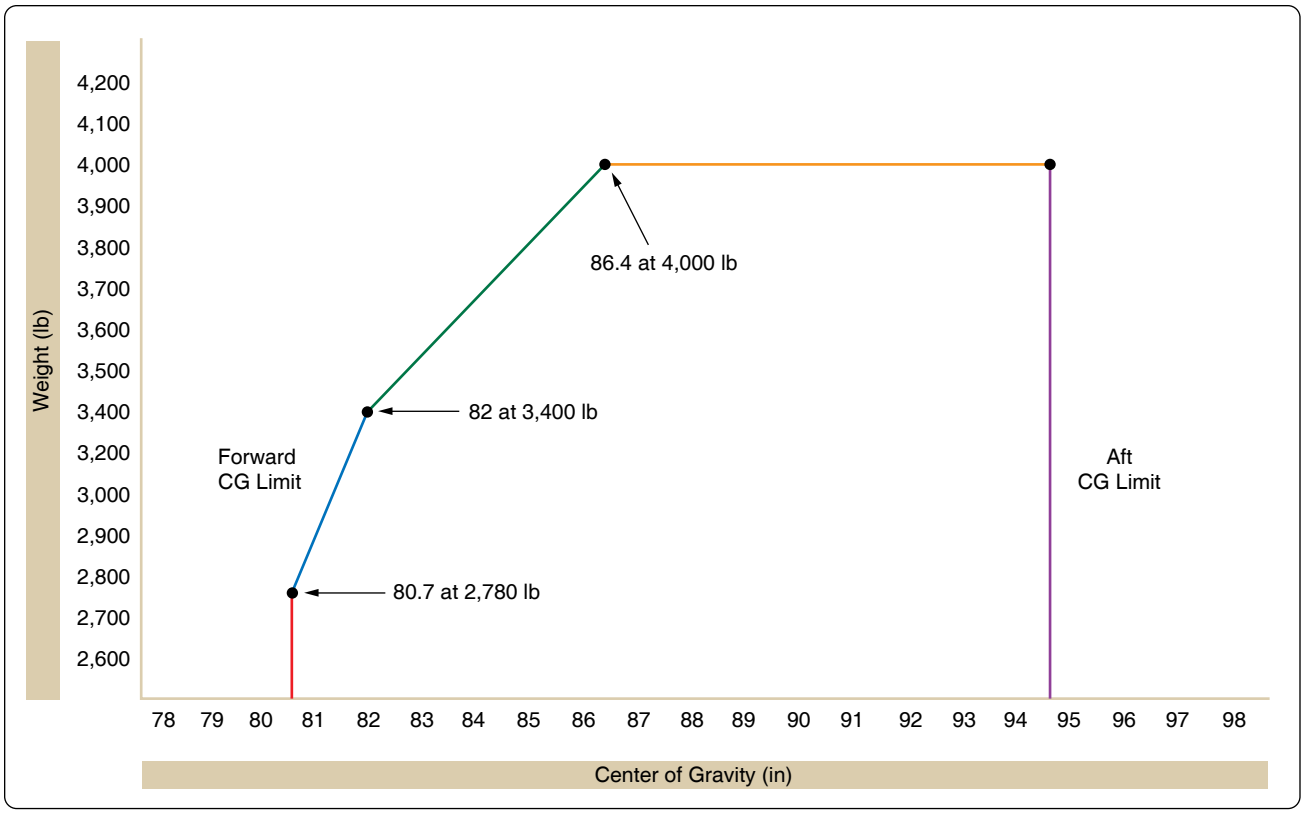


Figure 6-25. *Center of gravity envelope for the Piper Seneca.*

main-wheel weighing points. This is distance (D). The weight of the nosewheel (F) is 340 lb, and the distance between main wheels and nosewheel (L) is 78". The total weight of the airplane (W) is 2,006 lb.

The location of the CG may be determined by using this formula:

$$CG = D - \left(\frac{F \times L}{W} \right)$$

$$= 128 - \left(\frac{340 \times 78}{2{,}006} \right)$$

$$= 114.8$$

The CG is 114.8" aft of the datum. This is 13.2" forward of the main-wheel weighing points, which proves the location of the datum has no effect on the location of the CG so long as all measurements are made from the same location.

Datum Aft of the Main Wheels–Nosewheel Landing Gear

The datum of some aircraft may be located aft of the main wheels. The airplane in this example is the same one just discussed, but the datum is at the intersection of the trailing edge of the wing with the fuselage. The distance (D) between the datum of the airplane in *Figure 6-30* and the main-wheel weighing points is 75", the weight of the nosewheel (F) is 340 lb, and the distance between main wheels and nosewheel (L) is 78". The total net weight of the airplane (W) is 2,006 lb. The location of the CG may be determined by using this formula:

$$CG = -\left(D + \frac{F \times L}{W} \right)$$

$$= -\left(75 + \frac{340 \times 78}{2{,}006} \right)$$

$$= -88.2$$

The CG location is a negative value, which means it is 88.2" forward of the datum. This places it 13.2" forward of the main wheels, the same location as it was when it was measured from other datum locations.

Location of Datum

It makes no difference where the datum is located if all measurements are made from the same location.

Datum Forward of the Main Wheels–Tail Wheel

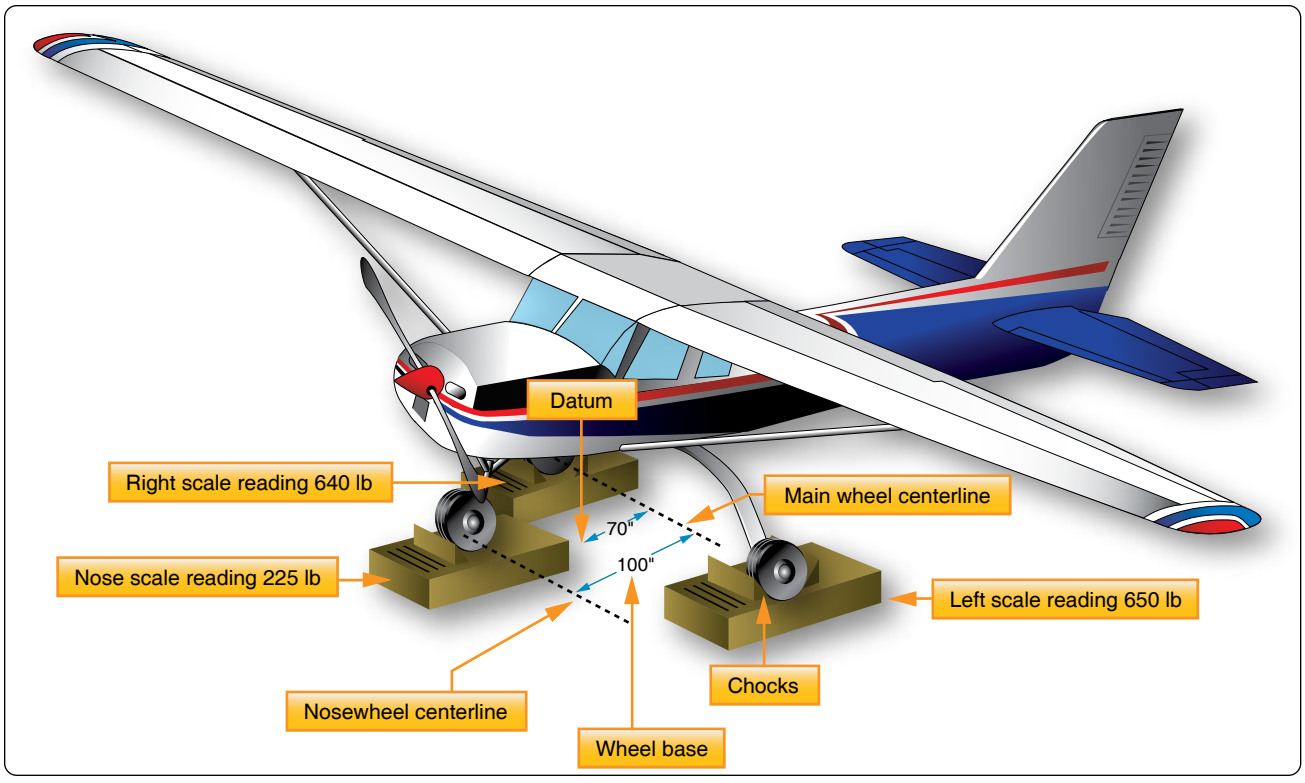


Figure 6-26. *Example airplane being weighed. The datum is 70" forward of the wing root leading edge.*

Aircraft datum:	Leading edge of the wing
Leveling means:	Two screws, left side of fuselage below window
Wheelbase:	100"
Fuel capacity:	30 gal aviation gasoline at +95"
Unusable fuel:	6 lb at +98"
Oil capacity:	8 qt at –38"
Note 1:	Empty weight includes unusable fuel and full oil
Left main scale reading:	650 lb
Right main scale reading:	640 lb
Nose scale reading:	225 lb
Tare weight:	5 lb chocks on left main 5 lb chocks on right main 2.5 lb chock on nose
During weighing:	Fuel tanks full and oil full Hydrometer check on Fuel shows 5.9 lb/gal

Figure 6-27. *Specifications and weighing specific data for tricycle gear airplane.*

Landing Gear

Locating the CG of a tail wheel airplane is done in the same way as locating it for a nosewheel airplane except the formulas use $\frac{R \times L}{W}$ rather than $\frac{F \times L}{W}$.

The distance (D) between the datum of the airplane in *Figure 6-31* and the main-gear weighing points is 7.5", the weight of the tail wheel (R) is 67 lb, and the distance (L) between the main-wheel and the tail wheel weighing points is 222". The total weight of the airplane (W) is 1,218 lb. Determine the CG by using this formula:

Item	Weight (lb)	Tare (lb)	Net Wt. (lb)	Arm (inches)	Moment (in-lb)
Nose	225	–2.5	222.5	–30	–6,675
Left Main	650	–5	645	+70	45,150
Right Main	640	–5	635	+70	44,450
Subtotal	1,515	–12.5	1,502.5		82,925
Fuel Total			–177	+95	–16,815
Fuel Unuse			+6	+98	588
Oil			Full		
Total			**1,331.5**	**+50.1**	**66,698**

Figure 6-28. *Center of gravity calculation for airplane being weighed.*

$$CG = D + \left(\frac{R \times L}{W} \right)$$

$$= 7.5 + \left(\frac{67 \times 222}{1{,}218} \right)$$

$$= 19.7$$

The CG is 19.7 inches behind the datum.

Datum Aft of the Main Wheels–Tail Wheel Landing Gear

The datum of the airplane in *Figure 6-32* is located at the intersection of the wing root trailing edge and the fuselage. This places the arm of the main gear (D) at –80". The net weight of the tail wheel (R) is 67 lb, the distance between the main wheels and the tail wheel (L) is 222", and the total net weight (W) of the airplane is 1,218 lb.

Since the datum is aft of the main wheels, use the formula:

$$CG = -D + \left(\frac{R \times L}{W} \right)$$

$$= -80 + \left(\frac{67 \times 222}{1{,}218} \right)$$

$$= -67.8$$

The CG is 67.8" forward of the datum, or 12.2" aft of the main-gear weighing points. The CG is in the same location relative to the main wheels, regardless of where the datum is located.

Loading an Aircraft for Flight

The ultimate test of whether there is a problem with an airplane's weight and balance is when it is loaded and ready to fly. The only real importance of an airplane's empty weight and EWCG is how it affects the loaded weight and balance of the airplane, since an airplane does not fly when it is empty. The pilot-in-command is responsible for the weight and balance of the loaded airplane, and they make the final decision on whether the airplane is safe to fly.

Example Loading of an Airplane

As an example of an airplane being loaded for flight, the Piper Seneca twin will be used. The TCDS for this airplane was shown earlier in this chapter, and its CG range and CG envelope were also shown.

The information from the TCDS that pertains to this example loading is shown in *Figure 6-33*.

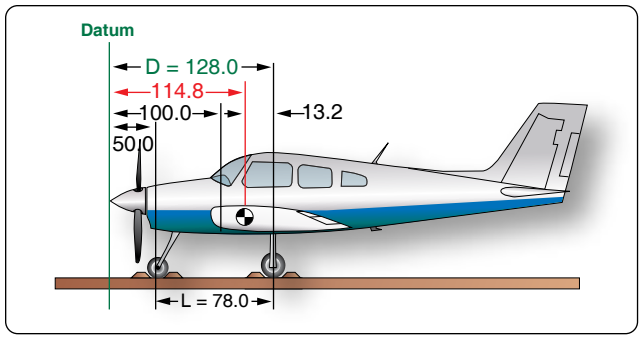


Figure 6-29. *The datum is 100" forward of the wing root leading edge.*

For the example loading of the airplane, the following information applies:

- Airplane Serial Number: 34-7250816
- Airplane Empty Weight: 2,650 lb
- Airplane EWCG: +86.8"

For today's flight, the following useful load items are included:

- 1 pilot at 180 lb at an arm of +85.5"
- 1 passenger at 160 lb at an arm of +118.1"
- 1 passenger at 210 lb at an arm of +118.1"
- 1 passenger at 190 lb at an arm of +118.1"
- 1 passenger at 205 lb at an arm of +155.7"
- 50 lb of baggage at an arm of +22.5"
- 100 lb of baggage at an arm of +178.7"
- 80 gal of fuel at an arm of +93.6"

To calculate the loaded weight and CG of this airplane, a four-column chart is used in *Figure 6-34*.

Based on the information in the TCDS, the maximum takeoff weight of this airplane is 4,200 lb and the aft-most CG limit

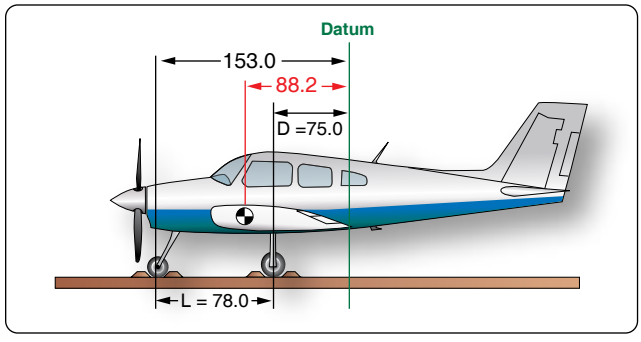


Figure 6-30. *The datum is aft of the main wheels at the wing trailing edge.*

is +94.6". The loaded airplane in *Figure 6-34* is 25 lb too heavy, and the CG is 1.82" too far aft. To make the airplane safe to fly, the load needs to be reduced by 25 lb and some of the load needs to be shifted forward. For example, the baggage can be reduced by 25 lb, and a full 100 lb of it can be placed in the more forward compartment. One passenger can be moved to the forward seat next to the pilot, and the aft-most passenger can then be moved forward.

With the changes made, the loaded weight is now at the maximum allowable of 4,200 lb, and the CG has moved forward 4.42". *[Figure 6-35]* The airplane is now safe to fly.

Adverse-Loaded CG Checks

Many modern aircraft have multiple rows of seats and often more than one baggage compartment. After any repair or alteration that changes the weight and balance, the A&P mechanic or repairman must ensure that no legal condition of loading can move the CG outside of its allowable limits. To determine this, adverse-loaded CG checks must be performed and the results noted in the weight and balance revision sheet.

During a forward adverse-loaded CG check, all useful load items in front of the forward CG limit are loaded and all useful load items behind the forward CG limit are left empty. So, if there are two seats and a baggage compartment located in front of the forward CG limit, two people weighing 170 lb each are seated and the maximum allowable baggage is placed in the baggage compartment. Any seat or baggage compartment located behind the forward CG limit is left empty. If the fuel is located behind the forward CG limit, minimum fuel will be shown in the tank. Minimum fuel is calculated by dividing the engine's METO hp by 2.

During an aft adverse-loaded CG check, all useful load items behind the aft CG limit are loaded and all useful load items in front of the aft CG limit are left empty. Even though the pilot's seat will be in front of the aft CG limit, the pilot's seat cannot be left empty. If the fuel tank is located forward of the aft CG limit, minimum fuel will be shown.

Example Forward & Aft Adverse-Loaded CG Checks

Using the stick airplane in *Figure 6-36* as an example, adverse forward and aft CG checks are calculated. Some of the data for the airplane is shown in *Figure 6-36*, such as seat, baggage, and fuel information. The CG limits are shown, with arrows pointing in the direction where maximum and minimum weights are loaded. On the forward check, any useful load item located in front of 89" is loaded, and anything behind that location is left empty. On the aft check, maximum weight is added behind 99" and minimum weight in front of that location. For either of the checks, if fuel is not located in a maximum weight location, minimum fuel must

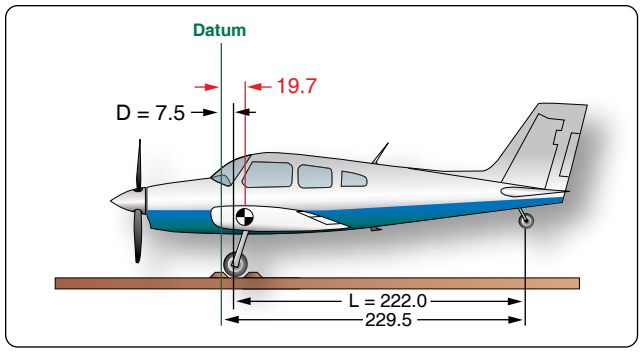


Figure 6-31. *The datum of this tail wheel airplane is the wing root leading edge.*

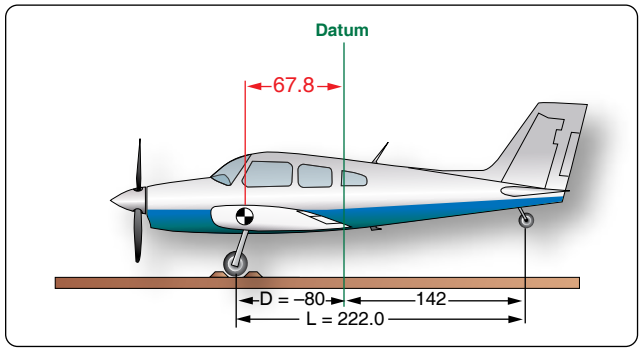


Figure 6-32. *The datum is aft of the main wheels, at the intersection of the wing trailing edge and the fuselage.*

be accounted for. Notice that the front seats show a location of 82" to 88", meaning they are adjustable fore and aft. In a forward check, the pilot's seat will be shown at 82", and in the aft check it will be at 88". Additional specifications for the airplane shown in *Figure 6-36* are as follows:

- Airplane empty weight: 1,850 lb
- EWCG: +92.45"
- CG limits: +89" to +99"
- Maximum weight: 3,200 lb
- Fuel capacity: 45 gal at +95" (44 usable)
40 gal at +102" (39 usable)

In evaluating the two extreme condition checks, the following key points should be recognized. *[Figure 6-37]*

- The total arm is the airplane CG and is found by dividing the total moment by the total weight.
- For the forward check, the only thing loaded behind the forward limit was minimum fuel.
- For the forward check, the pilot and passenger seats were shown at the forward position of 82".

CG Range (Gear Extended)	S/N 34-7250215 through 34-7450220: (+87.9") to (+94.6") at 4,200 lb (+82.0") to (+94.6") at 3,400 lb (+80.7") to (+94.6") at 2,780 lb Straight line variation between points given. –32 in-lb moment change due to gear retracting landing gear
Empty Weight CG Range	None
Maximum Weight	S/N 34-7250215 through 34-7450220: 4,200 lb—Takeoff 4,000 lb—Landing
No. of Seats	7 (2 at +85.5", 3 at +118.1", 2 at +155.7")
Maximum Baggage	200 lb (100 lb at +22.5, 100 lb at +178.7)
Fuel Capacity	98 gal (2 wing tanks) at (+93.6") (93 gal usable). See NOTE 1 for data on system fuel.

Figure 6-33. *Example loading information pertaining to TCDS.*

- For the forward check, the CG was within limits, so the airplane could be flown this way.
- For the aft check, the only thing loaded in front of the aft limit was the pilot, at an arm of 88".
- For the aft check, the fuel tank at 102" was filled, which more than accounted for the required minimum fuel.
- For the aft check, the CG was out of limits by 0.6", so the airplane should not be flown this way.

Equipment Change & Aircraft Alteration

When the equipment in an aircraft is changed, such as the installation of a new radar system or ground proximity warning system, or the removal of a radio or seat, the weight and balance of an aircraft changes. An alteration performed on an aircraft, such as a cargo door being installed or a reinforcing plate being attached to the spar of a wing, also changes the weight and balance of an aircraft. Any time the equipment is changed or an alteration is performed, the new empty weight and EWCG must be determined. This can be accomplished by placing the aircraft on scales and weighing it, or by mathematically calculating the new weight and balance. The mathematical calculation is acceptable if the exact weight and arm of all the changes are known.

Example Calculation After an Equipment Change

A small, twin-engine airplane has some new equipment installed and some of its existing equipment removed. The details of the equipment changes are shown in *Figure 6-38.* To calculate the new empty weight and EWCG, a four-column chart is used. *[Figure 6-39]* In evaluating the weight and balance calculation shown in *Figure 6-39,* the following key points should be recognized.

- The weight of the equipment needs to be identified with a plus or minus to signify whether it is being installed or removed.
- The sign of the moment (plus or minus) is determined by the signs of the weight and arm.
- The strobe and the ADF are both being removed (negative weight), but only the strobe has a negative moment. This is because the arm for the ADF is also negative, and two negatives multiplied together produce a positive result.
- The total arm is the airplane's CG and is found by dividing the total moment by the total weight.

Item	Weight (lb)	Arm (inches)	Moment (in-lb)
Empty Weight	2,650	+86.80	230,020.0
Pilot	180	+85.50	15,390.0
Passenger	160	+118.10	18,896.0
Passenger	210	+118.10	24,801.0
Passenger	190	+118.10	22,439.0
Passenger	205	+155.70	31,918.5
Baggage	50	+22.50	1,125.0
Baggage	100	+178.70	17,870.0
Fuel	480	+93.60	44,928.0
Total	**4,225**	**+96.42**	**407, 387.50**

Figure 6-34. *Center of gravity calculation for Piper Seneca.*

Item	Weight (lb)	Arm (inches)	Moment (in-lb)
Empty Weight	2,650	+86.80	230,020.0
Pilot	180	+85.50	15,390.0
Passenger	210	+85.50	17,955.0
Passenger	160	+155.01	24,801.6
Passenger	190	+118.10	22,439.0
Passenger	205	+118.10	24,210.5
Baggage	100	+22.50	2,250.0
Baggage	25	+178.70	4,467.5
Fuel	480	+93.60	44,928.0
Total	**4,200**	**+92.01**	**386,461.0**

Figure 6-35. *Center of gravity calculation for Piper Seneca with weights shifted.*

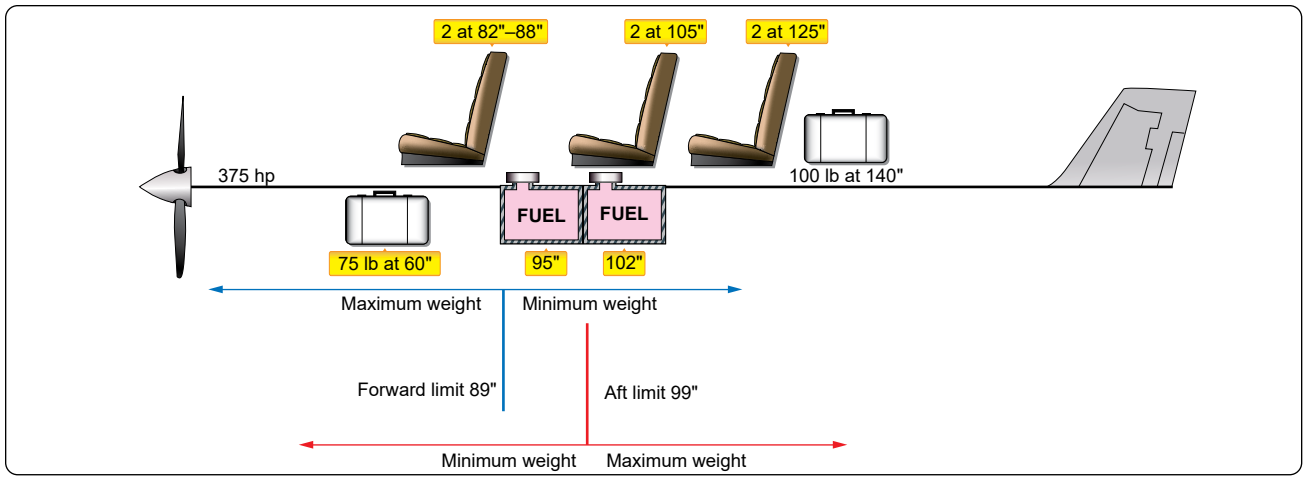


Figure 6-36. *Example airplane for extreme condition checks.*

- The result of the equipment change is that the airplane's weight was reduced by 22.5 lb and the CG has moved forward 0.67".

Use of Ballast

Ballast is used in an aircraft to attain the desired CG balance, when the CG is not within limits or is not at the location desired by the operator. It is usually located as far aft or as far forward as possible to bring the CG within limits, while using a minimum amount of weight.

Temporary Ballast

Temporary ballast, in the form of lead bars, heavy canvas bags of sand, or lead shot, is often carried in the baggage compartments to adjust the balance for certain flight conditions. The bags are marked "Ballast XX Pounds–Removal Requires Weight and Balance Check." Temporary ballast must be secured so it cannot shift its location in flight, and the structural limits of the baggage compartment must not be exceeded. All temporary ballast must be removed before the aircraft is weighed.

Temporary Ballast Formula

The CG of a loaded airplane can be moved into its allowable range by shifting passengers or cargo or by adding temporary ballast. To determine the amount of temporary ballast needed, use this formula:

$$\text{Ballast weight needed} = \frac{\text{Total wt.} \times \text{dist. needed to shift CG}}{\text{Dist. between ballast and desired CG}}$$

Figures 6-36 and *6-40* show an aft adverse-loaded CG check being performed on an airplane. In this previous example, the airplane's CG was out of limits by 0.6". If there were a need or a desire to fly the airplane loaded this way, one way to make it possible would be the installation of temporary ballast in the front of the airplane. The logical choice for placement of this ballast is the forward baggage compartment. The CG for this airplane is 0.6" too far aft. If the forward baggage compartment is used as a temporary ballast location, the ballast calculation will be as shown in *Figure 6-41*.

$$\text{Ballast weight needed} = \frac{\text{Total wt.} \times \text{dist. needed to shift CG}}{\text{Dist. between ballast and desired CG}}$$

Extreme Condition Forward Check			
Item	**Weight (lb)**	**Arm (inches)**	**Moment (in-lb)**
Empty Weight	1,850.0	+92.45	171,032.5
Pilot	170.0	+82.00	13,940.0
Passenger	170.0	+82.00	13,940.0
Baggage	75.0	+60.00	4,500.0
Fuel	187.5	+95.00	17,812.5
Total	**2,452.5**	**+90.20**	**221,225.0**

Extreme Condition Aft Check			
Item	**Weight (lb)**	**Arm (inches)**	**Moment (in-lb)**
Empty Weight	1,850	+92.45	171,032.5
Pilot	170	+88.00	14,960.0
2 Passengers	340	+105.00	35,700.0
2 Passengers	340	+125.00	42,500.0
Baggage	100	+140.00	14,000.0
Fuel	234	+102.00	23,868.0
Total	**3,034**	**+99.60**	**302,060.5**

Figure 6-37. *Center of gravity extreme conditions check.*

$$= \frac{3{,}034 \text{ lb} \times (0.6")}{39"}$$

$$= 46.68 \text{ lb}$$

When ballast is calculated, the answer should always be rounded up to the next higher whole pound, or in this case, 47 lb of ballast would be used. To ensure the ballast calculation is correct, the weight of the ballast should be plugged back into the four-column calculation and a new CG calculated.

The aft limit for the airplane was 99", and the new CG is at 98.96", which puts it within acceptable limits. The new CG did not fall exactly at 99" because the amount of needed ballast was rounded up to the next whole pound. If the ballast could have been placed farther forward, such as being bolted to the engine firewall, less ballast would have been needed. That is why ballast is always placed as far away from the affected limit as possible.

In evaluating the ballast calculation shown above, the following key points should be recognized.

- The loaded weight of the aircraft, as identified in the formula, is what the airplane weighed when the CG was out of limits.
- The distance the CG is out of limits is the difference between the CG location and the CG limit, in this case 99.6" minus 99".
- The affected limit identified in the formula is the CG limit which has been exceeded. If the CG is too far aft, it is the aft limit that has been exceeded.
- The aft limit for this example is 99", and the ballast is being placed in the baggage compartment at an arm of 60". The difference between the two is 39", the quantity divided by in the formula.

Viewed as a first-class lever problem, *Figure 6-42* shows what this ballast calculation would look like. A ballast weight of 46.68 lb on the left side of the lever multiplied by the arm of 39" (99 minus 60) would equal the aircraft weight of 3,034 lb multiplied by the distance the CG is out of limits, which is 0.6" (99.6 minus 99).

Permanent Ballast

If a repair or alteration causes the aircraft CG to fall outside of its limit, permanent ballast can be installed. Usually, permanent ballast is made of blocks of lead painted red and marked "Permanent Ballast–Do Not Remove." It should be attached to the structure so that it does not interfere with any control action, and attached rigidly enough that it cannot be dislodged by any flight maneuvers or rough landing. The installation of permanent ballast results in an increase in the aircraft empty weight, and it reduces the useful load.

Three things must be known to determine the amount of ballast needed to bring the CG within limits: the amount the CG is out of limits, the distance between the location of the ballast, and the limit that is affected. If an airplane with an empty weight of 1,876 lb has been altered so its EWCG is +32.2, and CG range for weights up to 2,250 lb is +33.0 to +46.0, permanent ballast must be installed to move the EWCG from +32.2 to +33.0. There is a bulkhead at fuselage station 228 strong enough to support the ballast. To determine the amount of ballast needed, use this formula:

$$\text{Ballast weight} = \frac{\text{Aircraft empty wt.} \times \text{dist. out of limits}}{\text{Dist. between ballast and desired CG}}$$

$$= \frac{1{,}876 \text{ lb} \times 0.8"}{}$$

Airplane empty weight:	2,350 lb
Airplane EWCG:	+24.7"
Airplane datum:	Leading edge of the wing
Radio installed:	5.8 lb at an arm of –28"
Global positioning system installed:	7.3 lb at an arm of –26"
Emergency locater transmitter installed:	2.8 lb at an arm of +105"
Strobe light removed:	1.4 lb at an arm of +75"
Automatic direction finder (ADF) removed:	3 lb at an arm of –28"
Seat removed:	34 lb at an arm of +60"

Figure 6-38. *Twin-engine airplane equipment changes.*

Item	Weight (lb)	Arm (inches)	Moment (in-lb)
Empty Weight	2,350.0	+24.70	58,045.0
Radio Install	+5.8	–28.00	–162.4
GPS Install	+7.3	–26.00	–189.8
ELT Install	+2.8	+105.00	294.0
Strobe Remove	–1.4	+75.00	–105.0
ADF Remove	–3.0	–28.00	84.0
Seat Remove	–34.0	+60.00	–2,040.0
Total	**2,327.5**	**24.03**	**55,925.8**

Figure 6-39. *Center of gravity calculation after equipment change.*

$$228 - 33$$

$$= \frac{1{,}500.8}{195}$$

$$= 7.7 \text{ lb}$$

A block of lead weighing 7.7 pounds attached to the bulkhead at fuselage station 228, moves the EWCG back to its proper forward limit of +33. This block should be painted red and marked "Permanent Ballast– Do Not Remove."

Loading Graphs & CG Envelopes

The weight and balance computation system, commonly called the loading graph and CG envelope system, is an excellent and rapid method for determining the CG location for various loading arrangements. This method can be applied to any make and model of aircraft, but is more often seen with small GA aircraft.

Aircraft manufacturers using this method of weight and balance computation prepare graphs like those shown in *Figures 6-43* and *6-44* for each make and model aircraft at the time of original certification. The graphs become a permanent part of the aircraft records and are typically found in the AFM/POH. These graphs, used in conjunction with the empty weight and EWCG data found in the weight and balance report, allow the pilot to plot the CG for the loaded aircraft.

The loading graph in *Figure 6-43* is used to determine the index number (moment value) of any item or weight that may be involved in loading the aircraft. To use this graph, find the point on the vertical scale that represents the known weight. Project a horizontal line to the point where it intersects the proper diagonal weight line (i.e., pilot, copilot, baggage). Where the horizontal line intersects the diagonal, project a vertical line downward to determine the loaded moment (index number) for the weight being added.

After the moment for each item of weight has been determined, all weights are added and all moments are added. The total weight and moment is then plotted on the CG envelope. *[Figure 6-44]* The total weight is plotted on the vertical scale of the graph, with a horizontal line projected out from that point. The total moment is plotted on the horizontal scale of the graph, with a vertical line projected up from that point. Where the horizontal and vertical plot lines intersect on the graph is the CG for the loaded aircraft. If the point where the plot lines intersect falls inside the CG envelope, the aircraft CG is within limits. In *Figure 6-44,* there are two CG envelopes, one for the aircraft in the Normal Category and one for the aircraft in the Utility Category.

Item	Weight (lb)	Arm (inches)	Moment (in-lb)
Empty Weight	1,850	+92.45	171,032.5
Pilot	170	+88.00	14,960.0
2 Passengers	340	+105.00	35,700.0
2 Passengers	340	+125.00	42,500.0
Baggage	100	+140.00	14,000.0
Fuel	234	+102.00	23,868.0
Total	**3,034**	**+99.60**	**302,060.5**

Figure 6-40. *Extreme condition check.*

Item	Weight (lb)	Arm (inches)	Moment (in-lb)
Loaded Weight	3,034	+99.60	302,060.5
Ballast	47	+60.00	2,820.0
Total	**3,081**	**+98.96**	**304,880.5**

Figure 6-41. *Ballast calculation.*

The loading graph and CG envelope shown in *Figures 6-43* and *6-44* are for an airplane with the following specifications and weight and balance data.

- Number of seats: 4
- Fuel capacity (usable): 38 gal of Avgas
- Oil capacity: 8 qt (included in empty weight)
- Baggage: 120 lb
- Empty weight: 1,400 lb
- EWCG: 38.5"
- Empty weight moment: 53,900 in-lb

An example of loading the airplane for flight and calculating the total loaded weight and the total loaded moment is shown

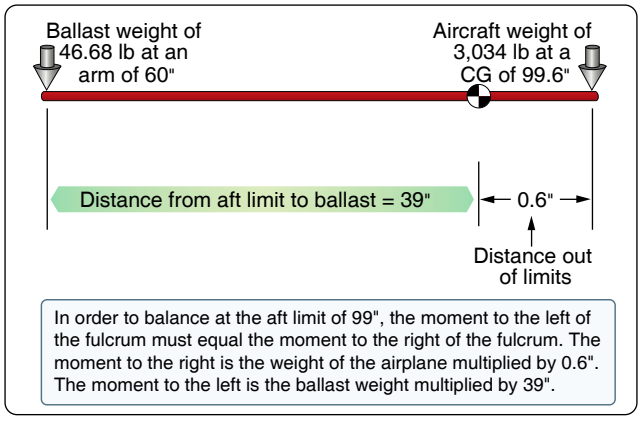


Figure 6-42. *Ballast calculation as a first class lever.*

in *Figures 6-45* and *6-46.* The use of the loading graph to determine the moment for each of the useful load items is shown in *Figure 6-46.* The color used for each useful load item in *Figure 6-45* matches the color used for the plot on the loading graph.

The total loaded weight of the airplane is 2,258 lb and the total loaded moment is 99,400 in-lb. These two numbers can now be plotted on the CG envelope to see if the airplane is within CG limits. *Figure 6-47* shows the CG envelope with the loaded weight and moment of the airplane plotted. The CG location shown falls within the normal category envelope, so the airplane is within CG limits for this category.

It is interesting to note that the lines that form the CG envelope are graphic plots of the forward and aft CG limits. In *Figure 6-47,* the red line is a graphic plot of the forward limit, and the blue and green lines are graphic plots of the aft limit for the two different categories.

Helicopter Weight & Balance

General Concepts

All the terminology and concepts that apply to airplane weight and balance also apply generally to helicopter weight and balance. However, there are some specific differences that need to be identified.

Most helicopters have a much more restricted CG range than airplanes. In some cases, this range is less than 3". The exact location and length of the CG range is specified for each helicopter and usually extends a short distance fore and aft of the main rotor mast or centered between the main rotors of a dual rotor system. Whereas airplanes have a CG range only along the longitudinal axis, helicopters have both longitudinal and lateral CG ranges. Because the wings extend outward from the CG, airplanes tend to have a great deal of lateral stability. A helicopter, on the other hand, acts like a pendulum, with the weight of the helicopter hanging from the main rotor shaft.

Ideally, the helicopter should have such perfect balance that the fuselage remains horizontal while in a hover. If the helicopter is too nose heavy or tail heavy while it is hovering, the cyclic pitch control is used to keep the fuselage horizontal. If the CG location is too extreme, it may not be possible to keep the fuselage horizontal or maintain control of the helicopter.

Helicopter Weighing

When a helicopter is being weighed, the location of both longitudinal and lateral weighing points must be known to determine its empty weight and EWCG. This is because helicopters have longitudinal and lateral CG limits. As with the airplane, the longitudinal arms are measured from the datum, with locations behind the datum being positive arms and locations in front of the datum being negative arms. Laterally, the arms are measured from the butt line, which is a line from the nose to the tail running through the middle of the helicopter. When facing forward, arms to the right of the butt line are positive; to the left they are negative.

Before a helicopter is weighed, it must be leveled longitudinally and laterally. This can be done with a spirit

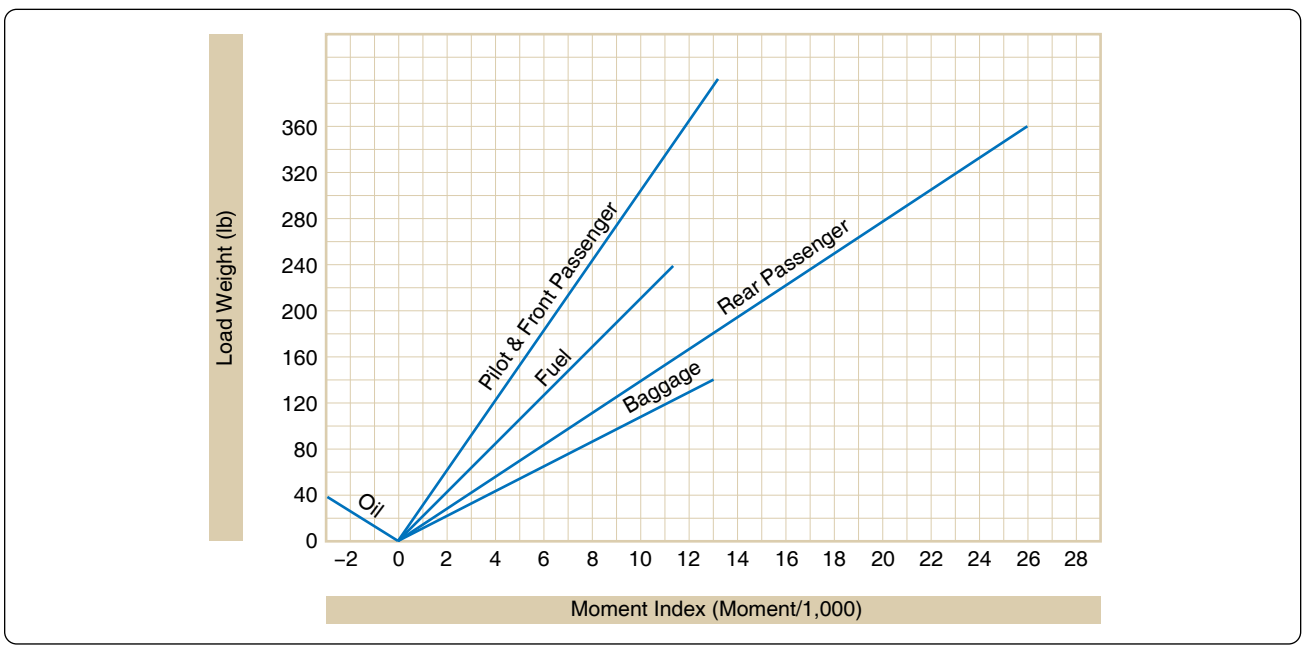


Figure 6-43. *Aircraft loading graph.*

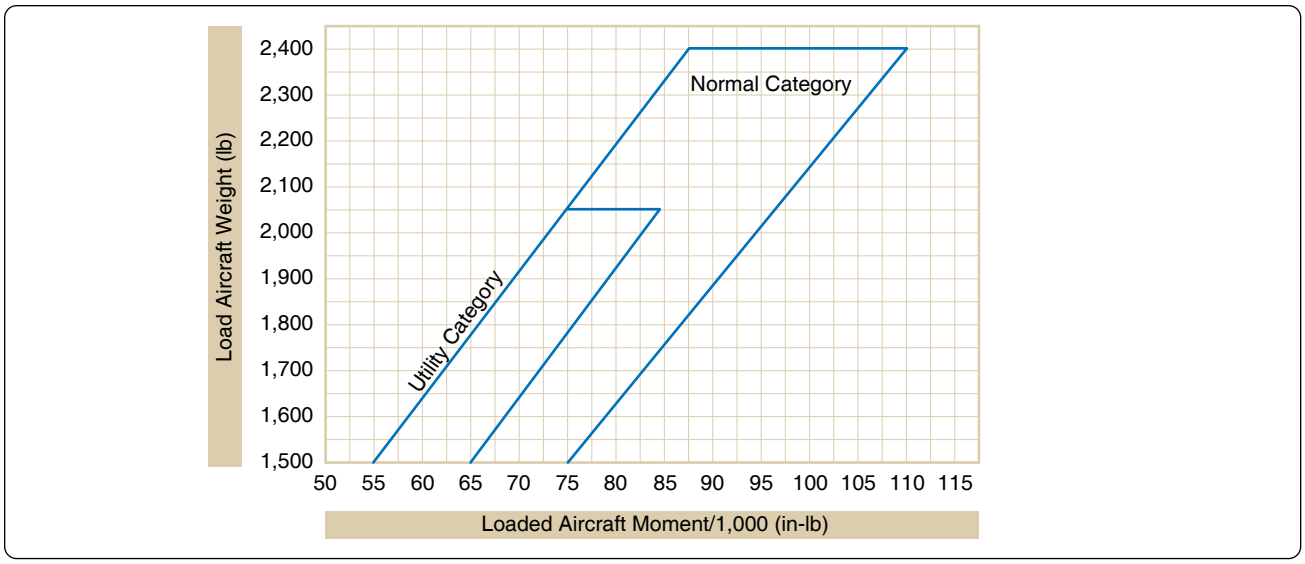


Figure 6-44. *CG envelope.*

level, but often it is done with a plumb bob. For example, the Bell JetRanger has a location inside the aft cabin where a plumb can be attached and allowed to hang down to the cabin floor. On the cabin floor is a plate bearing cross hairs that correspond to the horizontal and lateral axis of the helicopter. When the point of the plumb bob falls in the middle of the cross hairs, the helicopter is level along both axes. If the tip of the plumb bob falls forward of this point, the nose of the helicopter is too low; if it falls to the left of this point, the left side of the helicopter is too low. In other words, the tip of the plumb bob always moves toward the low point.

A Bell JetRanger helicopter is shown in *Figure 6-48* with the leveling plate depicted on the bottom right of the figure. The helicopter has three jack pads, two at the front and one in the back. To weigh this helicopter, three jacks would be placed on floor scales, and the helicopter would be raised off the hangar floor. To level the helicopter, the jacks would be adjusted until the plumb bob point falls exactly in the middle of the cross hairs.

Item	Weight (lb)	Moment (in-lb)
Aircraft Empty Weight	1,400	53,900
Pilot	180	6,000
Front Passengers	140	4,500
Rear Passengers	210	15,000
Baggage	100	9,200
Fuel	228	10,800
Total	**2,258**	**99,400**

Figure 6-45. *Aircraft load chart.*

As an example of weighing a helicopter, consider the Bell JetRanger in *Figure 6-48*, and the following specifications and weighing data shown in *Figure 6-49*.

Using six-column charts for the calculations, the empty weight and the longitudinal and lateral CG for the helicopter is shown in *Figure 6-50*. Based on the calculations in *Figure 6-50,* it has been determined that the empty weight of the helicopter is 1,985 lb, the longitudinal CG is at +108.73", and the lateral CG is at –0.31".

Weight and Balance—Weight-Shift Control Aircraft and Powered Parachutes

The terminology, theory, and concepts of weight and balance that applies to airplanes also applies to weight-shift control aircraft and powered parachutes. Weight is still weight, and the balance point is still the balance point. However, there are a few differences that need to be discussed. Before reading about the specifics of weight and balance on weight-shift control aircraft and powered parachutes, be sure to read about their aerodynamic characteristics in Chapter 5, Physics. Weight-shift control aircraft and powered parachutes do not fall under the same Code of Federal Regulations that govern certified airplanes and helicopters and, therefore, do not have TCDS or the same type of FAA-mandated weight and balance reports. Weight and balance information and guidelines are left to the individual owners and the companies with which they work in acquiring this type of aircraft. Overall, the industry that is supplying these aircraft is regulating itself well, and the safety record is good for those aircraft being operated by experienced pilots.

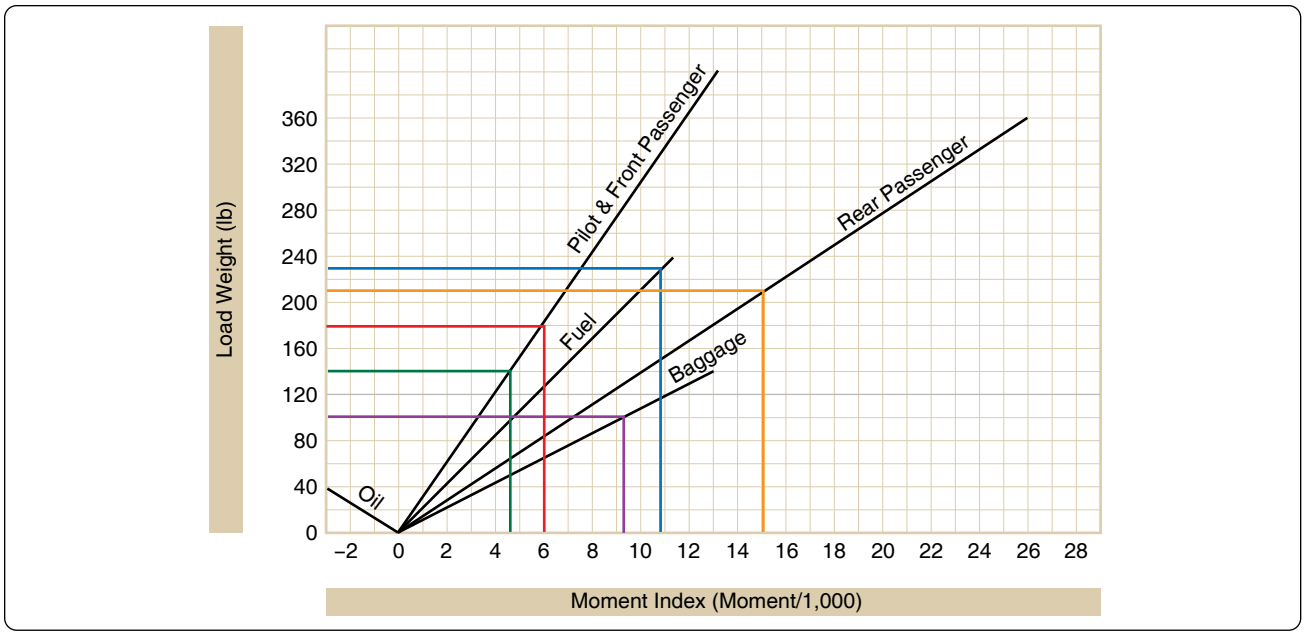


Figure 6-46. *Example plots on a loading graph.*

Weight-Shift Control Aircraft

Weight-shift control aircraft, commonly known by the name "trikes," have very few options for loading, because they have very few places to put useful load items. Some trikes have only one seat and a fuel tank, so the only variables for a flight are amount of fuel and weight of the pilot. Some trikes have two seats and a small storage bin in addition to the fuel tank.

The most significant factor affecting the weight and balance of a trike is the weight of the pilot; if the aircraft has two seats, the weight of the passenger must be considered. The trike acts somewhat like a single main rotor helicopter because the weight of the aircraft is hanging like a pendulum under the wing. *Figure 6-51* shows a two-place trike, in which the mast and the nose strut come together slightly below the wing attach point. When the trike is in flight, the weight of the aircraft is hanging from the wing attach point. The weight of the engine and fuel is behind this point, the passenger is almost directly below this point, and the pilot is forward of this point. The balance of the aircraft is determined by how all these weights compare.

The wing attach point, with respect to the wing keel, is an adjustable location. The attach point can be loosened and

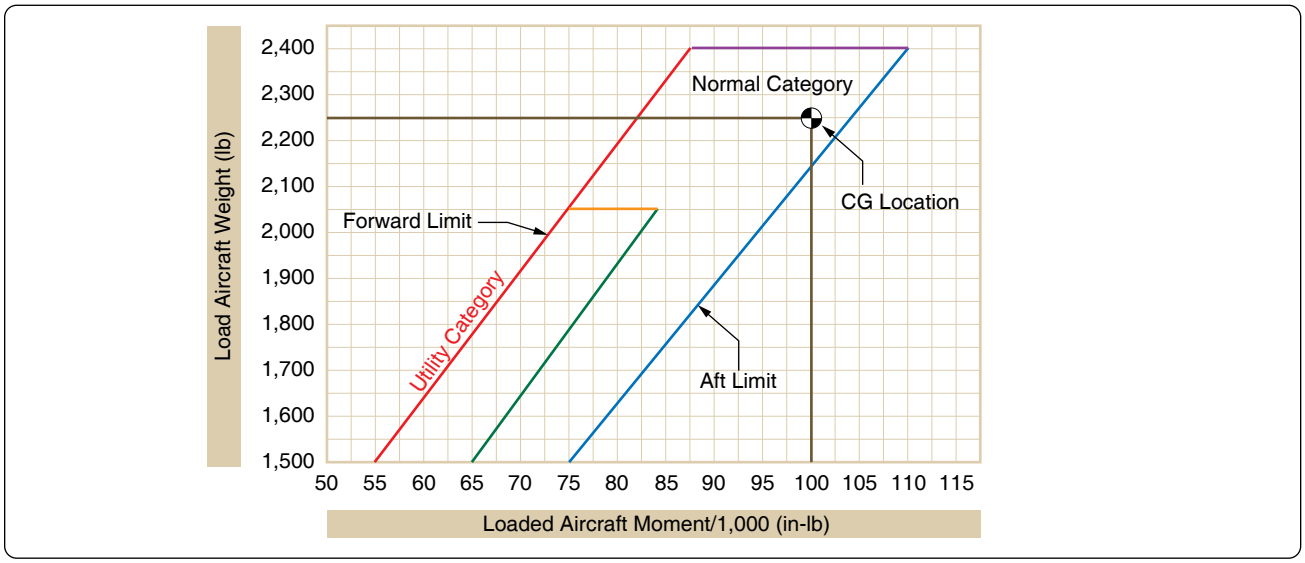


Figure 6-47. *CG envelope example plot.*

moved slightly forward or slightly aft, depending on the weight of the occupants. For example, if the aircraft is flown by a person that weighs more, the attach point can be moved a little farther aft, bringing the wing forward, to compensate for the change in CG. *Figure 6-52* shows a close-up of the wing attach point, and the small amount of forward and aft movement that is available.

Powered Parachutes

Powered parachutes have many of the same characteristics as weight-shift aircraft when it comes to weight and balance. They have the same limited loading, with only one or two seats and a fuel tank. They also act like a pendulum, with the weight of the aircraft hanging beneath the inflated wing (parachute).

The point at which the inflated wing attaches to the structure of the aircraft is adjustable to compensate for pilots and passengers of varying weights. With a very heavy pilot, the wing attach point would be moved forward to prevent the aircraft from being too nose heavy. *Figure 6-53* shows the structure of a powered parachute with the adjustable wing attach points.

Weight & Balance for Large Airplanes

Weight and balance for large airplanes is almost identical to what it is for small airplanes, on a much larger scale. If a technician can weigh a small airplane and calculate its empty weight and EWCG, that same technician should be able to do it for a large airplane. The jacks and scales are larger, and it may take more personnel to handle the equipment, but the concepts and processes are the same.

Built-In Electronic Weighing

One difference that may be found with large airplanes is the incorporation of electronic load cells in the aircraft's landing gear. With this type of system, the airplane can weigh itself as it sits on the tarmac. The load cells are built into the axles of the landing gear, or the landing gear strut, and they work in the same manner as load cells used with jacks. This system is currently in use on the Boeing 747-400, Boeing 777, Boeing 787, McDonnell Douglas MD-11, and the wide body Airbus airplanes like the A-330, A-340, and A-380.

The Boeing 777 utilizes two independent systems that provide information to the airplane's flight management system (FMS). If the two systems agree on the weight and CG of the airplane, the data being provided are considered accurate and the airplane can be dispatched based on that information. The flight crew has access to the information on the flight deck by accessing the FMS and bringing up the weight and balance page.

Mean Aerodynamic Chord

On small airplanes and on all helicopters, the CG location is identified as being a specific number of inches from the datum. The CG range is identified the same way. On larger airplanes, from private business jets to large jumbo jets, the CG and its range are typically identified in relation to the width of the wing.

The width of the wing on an airplane is known as the chord. If the leading edge and trailing edge of a wing are parallel to each other, the chord of the wing is the same along the wing's length. Business jets and commercial transport airplanes have wings that are tapered and that are swept back, so the width of their wings is different along their entire length. The width is greatest where the wing meets the fuselage and progressively decreases toward the tip. In relation to the aerodynamics of the wing, the average length of the chord on these tapered swept-back wings is known as the mean aerodynamic chord (MAC).

On these larger airplanes, the CG is identified as being at a location that is a specific percent of the mean aerodynamic chord (% MAC). For example, imagine that the MAC on an airplane is 100", and the CG falls 20" behind the leading edge of the MAC. That means it falls one-fifth of the way back, or at 20 percent of the MAC.

Figure 6-54 shows a large twin-engine commercial transport airplane. The datum is forward of the nose of the airplane, and all the arms are being measured from that point. The CG for the airplane is shown as an arm measured in inches. In the lower left corner of the figure, a cross section of the wing is shown, with the same CG information being presented.

To convert the CG location from inches to a percent of MAC, for the airplane shown in *Figure 6-54,* the steps are as follows:

1. Identify the CG location, in inches from the datum.
2. Identify the leading edge of the MAC (LEMAC), in inches from the datum.
3. Subtract LEMAC from the CG location.
4. Divide the difference by the length of the MAC.
5. Convert the result in decimals to a percentage by multiplying by 100.

As a formula, the solution to solve for the percent of MAC would be:

$$\text{Percent of MAC} = \frac{\text{CG} - \text{LEMAC}}{\text{MAC}} \times 100$$

The result using the numbers shown in *Figure 6-52* would be:

$$\text{Percent of MAC} = \frac{\text{CG} - \text{LEMAC}}{\text{MAC}} \times 100$$

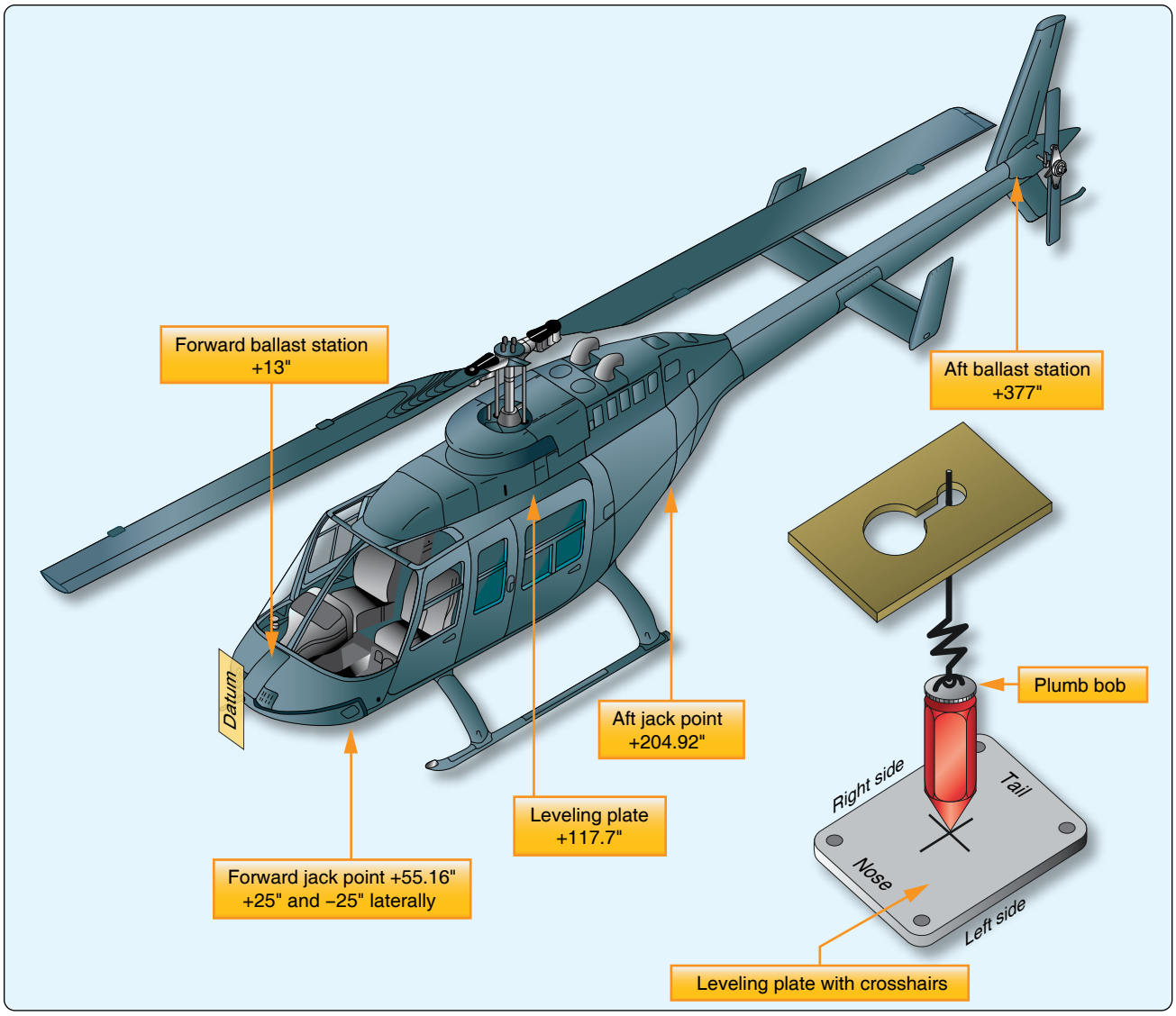


Figure 6-48. *Bell JetRanger.*

Datum:	55.16" forward of the front jack point centerline	**Left Front Scale Reading:**	650 lb
Leveling Means:	Plumb line from ceiling left rear cabin to index plate on floor	**Left Front Jack Point:**	Longitudinal arm of +55.16" Lateral arm of –25"
Longitudinal CG Limits:	+106" to +111.4" at 3,200 lb +106" to +112.1" at 3,000 lb +106" to +112.4" at 2,900 lb +106" to +113.4" at 2,600 lb +106" to +114.2" at 2,350 lb +106" to +114.2" at 2,100 lb Straight line variation between points	**Right Front Scale Reading:**	625 lb
		Right Front Jack Point:	Longitudinal arm of +55.16" Lateral arm of +25"
		After Scales Reading:	710 lb
Lateral CG Limits:	2.3" left to 3.0" right at longitudinal CG + 106.0" 3.0" left to 4.0" right at longitudinal CG +108" to +114.2" Straight line variation between points	**Aft Jack Point:**	Longitudinal arm of +204.92" Lateral arm of 0.0"
Fuel and Oil:	Empty weight includes unusable fuel and unusable oil	**Notes:**	The helicopter was weighed with unusable fuel and oil. Electronic scales were used and zeroed with the jacks in place, so no tare weight needs to be accounted for.

Figure 6-49. *Specifications and weighing data for Bell JetRanger.*

Longitudinal CG Calculation					
Item	Scale (lb)	Tare Wt. (lb)	Nt. Wt. (lb)	Arm (inches)	Moment (in-lb)
Left Front	650	0	650	+55.16	35,854.0
Right Front	625	0	625	+55.16	34,475.0
Aft	710	0	710	+204.92	145,493.2
Total	**1,985**		**1,985**	**+108.73**	**215,822.2**
Lateral CG Calculation					
Item	Scale (lb)	Tare Wt. (lb)	Nt. Wt. (lb)	Arm (inches)	Moment (in-lb)
Left Front	650	0	650	−25	−16,250
Right Front	625	0	625	+25	+15,625
Aft	710	0	710	0	0
Total	**1,985**		**1,985**	**+.31**	**−625**

Figure 6-50. *Center of gravity calculation for Bell JetRanger.*



Figure 6-51. *Weight and balance for a weight-shift control aircraft.*



Figure 6-52. *Wing attach point for a weight-shift control aircraft.*

$$= \frac{945 - 900}{\quad} \times 100$$



Figure 6-53. *Powered parachute structure with wing attach points.*

$$180$$

$$= 25 \text{ percent}$$

If the CG is known in percent of MAC, and there is a need to know the CG location in inches from the datum, the conversion would be done as follows:

1. Convert the percent of MAC to a decimal by dividing by 100.
2. Multiply the decimal by the length of the MAC.

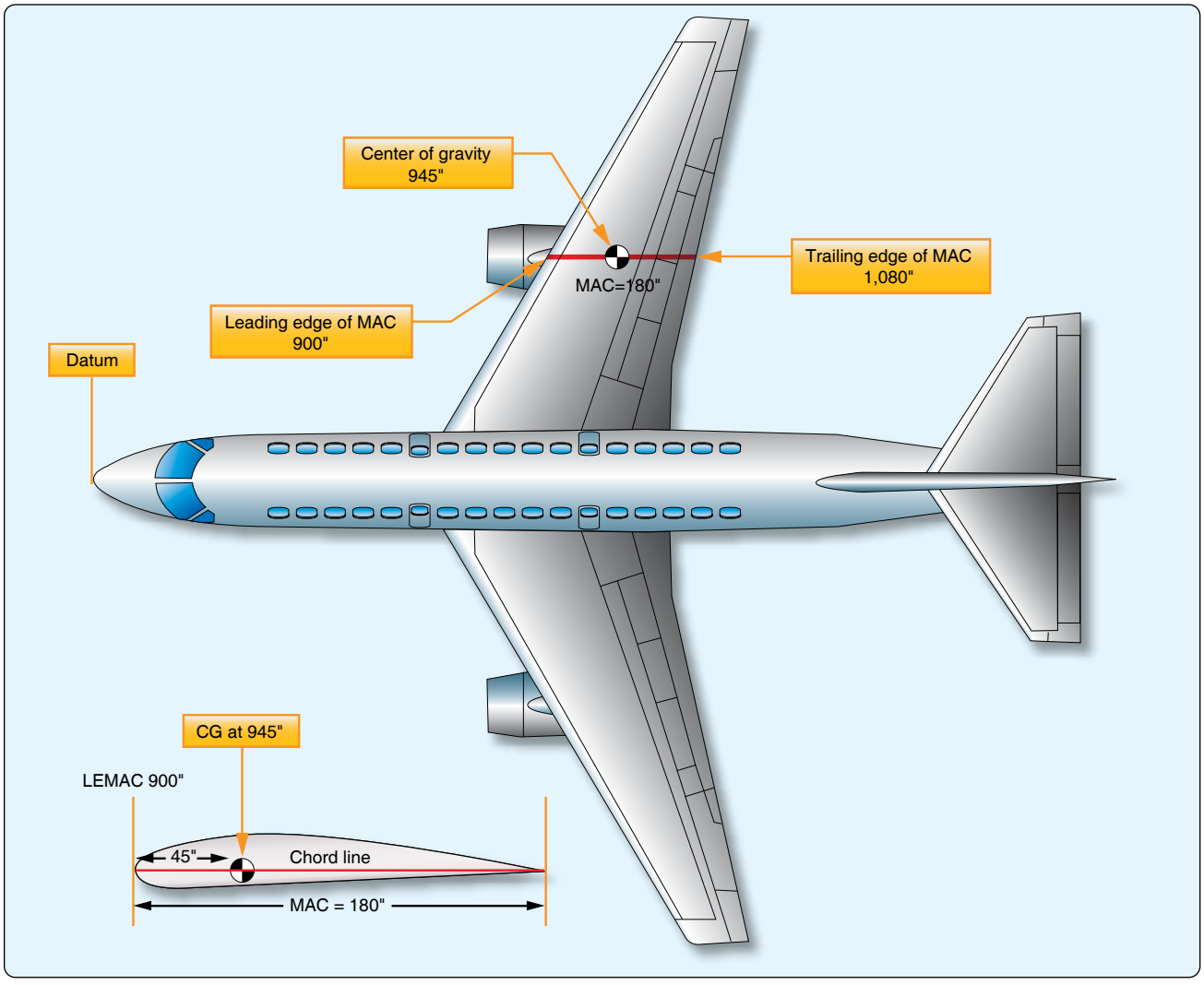


Figure 6-54. *Center of gravity location on a large commercial transport.*

3. Add this number to LEMAC.

As a formula, the solution to convert a percent of MAC to an inch value would be:

$$\text{CG in inches} = \%\,\text{MAC} \div 100 \times \text{MAC} + \text{LEMAC}$$

For the airplane in *Figure 6-54,* if the CG was at 32.5 percent of the MAC, the solution would be:

$$\begin{aligned}\text{CG in inches} &= \%\,\text{MAC} \div 100 \times \text{MAC} + \text{LEMAC}\\ &= 32.5 \div 100 \times 180 + 900\\ &= 958.5\end{aligned}$$

Weight & Balance Records

When a technician gets involved with the weight and balance of an aircraft, it almost always involves a calculation of the aircraft's empty weight and EWCG. Only on rare occasions are technicians involved in calculating adverse-loading CG checks, how much ballast is needed, or the loaded weight and balance of the aircraft. Calculating the empty weight and EWCG might involve putting the aircraft on scales and weighing it, or a pencil and paper exercise after installing a new piece of equipment.

The FAA requires that a current and accurate empty weight and EWCG be known for an aircraft. This information must be included in the weight and balance report, which is a part of the aircraft permanent records. The weight and balance report must be in the aircraft when it is being flown.

There is no required format for this report, but *Figure 6-55* is a good example of recording the data obtained from weighing an aircraft. As it is currently laid out, the form would accommodate either a tricycle gear or tail dragger airplane. Depending on the gear type, either the nose or the tail row would be used. If an airplane is being weighed using jacks and load cells, or if a helicopter is being weighed, the

Aircraft Weight and Balance Report

Results of Aircraft Weighing

Make _____ Model _____

Serial # _____ N# _____

Datum Location _____

Leveling Means _____

Scale Arms: Nose _____ Tail _____ Left Main _____ Right Main _____

Scale Weights: Nose _____ Tail _____ Left Main _____ Right Main _____

Tare Weights: Nose _____ Tail _____ Left Main _____ Right Main _____

Weight and Balance Calculation

Item	Scale (lb)	Tare Wt. (lb)	Net Wt. (lb)	Arm (inches)	Moment (in-lb)
Nose					
Tail					
Left Main					
Right Main					
Subtotal					
Fuel					
Oil					
Misc.					
Total					

Aircraft Current Empty Weight: _____

Aircraft Current Empty Weight CG: _____

Aircraft Maximum Weight: _____

Aircraft Useful Load: _____

Computed By: _____ (print name)

_____ (signature)

Certificate #: _____ (A&P, Repair Station, etc.)

Date: _____

Figure 6-55. *Aircraft weight and balance report.*

item names must be changed to reflect the weight locations.

If an equipment change is being done on an aircraft, and the new weight and balance is calculated mathematically instead of weighing the aircraft, the same type of form shown in *Figure 6-55* can be used. The only change would be the use of a four-column solution, instead of six columns, and there would be no tare weight or involvement with fuel and oil.